Predictive learning folds spatial maps along symmetries

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Place cells fire at the same spot in identical rooms set side by side, but much less when the rooms are rotated relative to one another, and lesions of the head-direction system bring the repetition back. Why does head direction keep rotated rooms apart? It tells the animal which way it faces, and it reads differently at matching spots in rotated rooms; a lesion removes both. We separated the two in recurrent networks trained only to predict their next view, which develop place cells and replay. Each network received one of two heading signals that carry one bit each about the current heading: one reads the same after a half-turn, the other flips. In an arena with a half-turn symmetry, networks given the flipping signal told each position from the position a half-turn away almost perfectly (decoding accuracy 0.95); networks given the unchanged signal could not (0.50, chance). What matters is not how much the signal says about heading but whether the symmetry changes it. Where it does not, the network gives one state to all the positions the symmetry relates, as if the arena had been folded onto itself, and we prove that, under the conditions met here, no learner could do otherwise. The fold constrains what can be learned from the map: a goal learned at one position transfers in full to the positions folded onto it, and rooms that share a code cannot be given different values. Silencing the heading signal in trained networks reproduces the direction of the lesion effects in rats. Single-cell symmetry scores can be high without any fold, so a fold must be tested in the population.

predictive learning | cognitive map | hippocampus | symmetry | place cells | head direction

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Author summary

Imagine waking up in one of two identical rooms. Looking around will not tell you which room you are in. A compass might, but only if the rooms face different ways: if one room is a shifted copy of the other, the compass reads the same at the matching spot in each. Rats seem to face this problem. Their place cells, which signal where the animal is, often fire at the same spot in identical rooms set side by side, much less when the rooms are turned relative to each other, and damage to the brain's compass brings the repetition back. Is it how much the compass says about direction that keeps turned rooms apart, or whether it reads differently in the two rooms? We trained artificial networks that develop place cells when they learn to predict what they will see next, and gave them two compass signals that carry the same amount of information but behave differently when the room is turned. Only the behaviour under turning mattered. Where a signal cannot tell two places apart, the network's map folds them into one, and anything learned at one place spreads to the other.

Introduction

Put a rat in a row of identical boxes and many of its place cells fire at the same spot in every box (1). The hippocampus, it seems, cannot tell the boxes apart. Now turn the boxes so that each faces a different way, and much of the repetition goes. When Grieves et al. (2) set four identical compartments at $60°$ to one another instead of side by side, place cells repeated their fields much less, and rats learned to tell the compartments apart more readily. Lesions point to the head-direction system: with the lateral mammillary nuclei destroyed, repetition across side-by-side compartments was unchanged, but across rotated compartments it came back (3), and fewer lesioned rats learned to tell rotated compartments apart (4).

But what is it about a head-direction signal that keeps rotated rooms apart? The signal has two properties, and either could do the job. It tells the animal which way it faces. And, because one room is turned relative to the other, it reads differently at matching spots in the two rooms. The usual account credits the first property, directional information (2, 3, 5). A lesion cannot decide, because it removes both properties at once. Yet the two predict different things: if information counts, any impoverished heading signal should blur the rooms together, and if the reading at matching spots counts, a single bit could still keep them apart.

A thought experiment suggests which. Suppose you wake up in one of two identical rooms and must say which one. Looking around cannot help: every spot in one room looks exactly like the matching spot in the other. A compass can help, but only sometimes. If the second room is a shifted copy of the first, the compass reads the same at matching spots, and you are stuck. If the second room is turned around, the compass reads differently, and you can tell. What matters is not how finely the compass is graduated but whether its reading differs between the two spots. The same holds inside one arena whose landmarks have a symmetry. Call two positions *twins* if a symmetry of the layout carries one onto the other: a half-turn gives each position one twin, the turns of a fourfold arena give it three, and in two identical rooms the twin of a spot is the same spot in the other room. In an arena with no symmetry we keep the name for positions a half-turn apart, as a control. If twins give the same views and the same self-motion signals, an agent that learns from those signals cannot tell them apart, whatever it learns. This is perceptual aliasing (6), and its result is a map that behaves as if the arena had been

folded onto itself. We call this a *fold*.

In a network, the two properties of the signal can be pulled apart. We build on sequential predictive learning (7), in which recurrent networks trained only to predict their upcoming views develop place cells and replay, and form a map of space only when they also receive speed and head direction. We ask not whether the head-direction input is needed, but what it decides about the map. Our agent walks a grid arena whose landmarks have no symmetry, a half-turn symmetry or a quarter-turn symmetry (the asymmetric, half-turn and quarter-turn arenas), or a maze of identical rooms modelled on the rat experiments. Besides its view and speed, each network receives one of four head-direction signals. *full* gives the heading and *const* gives nothing. *axis* reports whether the agent faces along the east–west or the north–south axis, so a half-turn leaves it unchanged. *flip* reports whether it faces east or south rather than west or north, so a half-turn flips it. These two carry exactly one bit each about the current heading, and differ in what a half-turn does to them. They also differ in how often a turn changes them, and the asymmetric arena controls for that. No animal can be given this pair of signals with everything else held fixed.

In the half-turn arena, networks given *flip* tell a position from its twin almost as well as networks given the full heading, while networks given *axis* cannot tell them apart at all, although the two signals perform identically in the asymmetric arena. One bit does the work of two when the symmetry flips it, and nothing when it does not. The folded code is still an accurate map of the folded arena, and it decides what can be learned from it: rooms that share a code cannot be given different values. Silencing the head-direction signal folds the map only where the arena has a symmetry, in the direction of the rat lesion results, and single cells can look symmetric where nothing is folded.

Results

A network that only predicts its views keeps translated rooms together and rotated rooms apart. Before explaining anything, we asked whether the model shows the animal result at all. We built two closed 6×6 rooms joined by a corridor, with identical landmarks inside, so that every view in one room also occurs, at the matching spot and heading, in the other (Methods). In one maze the second room is a translated copy of the first; in the other it is turned by a half-turn. Eight networks were trained on each maze with the full head-direction signal, and nothing in their training mentioned rooms.

The two mazes produced opposite maps (Fig. 2a). In translated rooms, units fired at the same spot in both rooms: the repetition score, the median correlation of a unit's firing map between the rooms, was 0.997. In rotated rooms the fields did not repeat (repetition -0.080), and a decoder tested at positions it had not been trained on read out the room at 0.987. This is not because one maze lost its map. Position within a room was decodable in both mazes, and every translation network differed from every rotation network on all of these measures (Methods). It is the contrast that Grieves et al. (2) recorded in CA1.

Pooled over all room steps, a decoder tested on paths it never saw in training reads the translated room at 0.497, which is chance (replicate set, eight networks per maze; Methods). That number mixes two different situations. On most room steps no learner could know the room, and on the rest the networks know it for a while and then lose it; we separate the two below, in "Beyond the forced fold". Four identical rooms in a row, as in the experiment of Spiers et al. (1), give the same picture: fields repeat across all four (repetition 0.994; Fig. 2b), and with whole paths held out, in a replicate set, the room decodes at 0.238 against a chance of 0.25 (Methods).

Why twins must share a state. The translated rooms need no trained network to explain them, which is the point of the thought experiment in the Introduction. A path through one room and the matching path through the other produce, step for step, the same views and the same movement signals. A network is a fixed function of its inputs, so from the same initial state it passes through the same states on both paths, and nothing that reads those states can say which room the agent is in. The argument is exact for an arena whose whole layout is symmetric, where it becomes a theorem.

Theorem. *Suppose the arena's landmarks are mapped onto themselves by a set of rotations G (the half-turn, or all four quarter-turns), that the agent is as likely to start at a position as at any of its twins, that it moves in the same way everywhere, and that the head-direction signal is unchanged by every rotation in G. Then the network's state cannot depend on which twin the agent occupies, and a decoder that must name the twin is right with probability at most $1/|G|$, that is, $1/2$ for a half-turn and $1/4$ for the quarter-turns.* The proof is in Methods; it specialises the belief invariance of group-invariant partially observable processes (8, Lemma 1) to a group under which every input is unchanged. Its conditions hold exactly in the open arenas, where the agent follows a random walk with the same action probabilities at every position and heading; the room mazes meet them only approximately, because the corridor between the rooms breaks the symmetry (Methods). They would fail for an animal that explored one room more than the other, because its own movements would then carry the room's identity.

The theorem is only an upper bound: it says which positions no learner can separate, not whether a given network will separate the others. A network might fail to, for two reasons. Its training objective might not require it: a network could predict its next view well without knowing which of two twins it is at. Or its head-direction signal might carry too little information, which is the account that the matched pair below is built to test. Whether a network separates the twins its inputs do distinguish is therefore an empirical question, and most of what follows is about it.

The theorem also says how to measure a fold. We train a linear decoder to name which of two twins the agent occupies, and call its accuracy the *disambiguation accuracy*: 0.5 is chance, and a fold pins it there. The decoder must not win by recognising a path it has already seen, so every number in the main text holds whole trajectories and whole sets

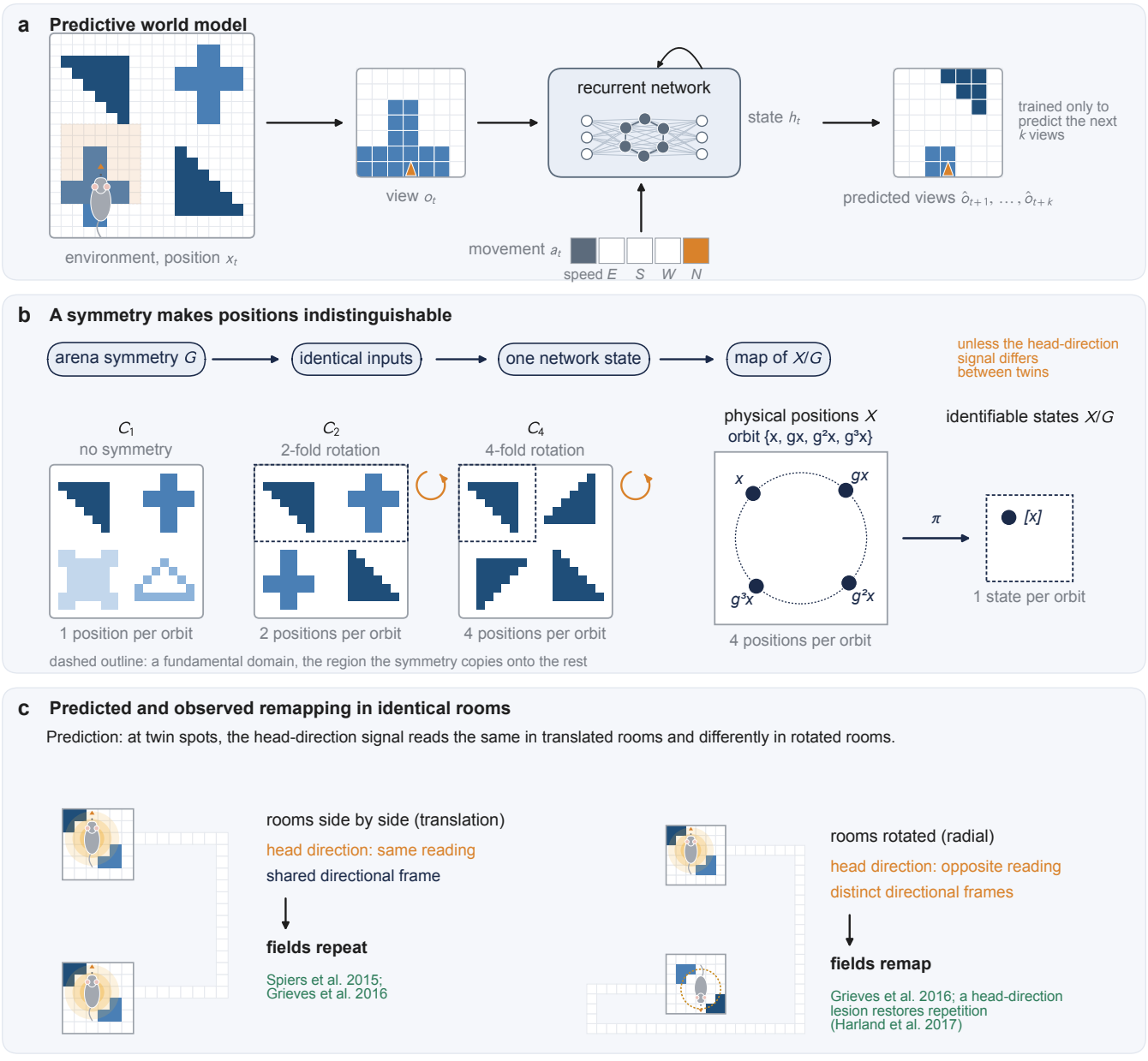


Fig. 1. Overview. (**a**) The model. An agent at position x_t in an arena (here the half-turn arena) sees a 7×7 egocentric view o_t (shaded square; the rat marks the agent, its orange tip the heading, and the triangle in each view marks the agent's own cell). A recurrent network receives the view and a five-dimensional movement input a_t, its speed and heading (Methods), and is trained only to predict the next k views. (**b**) The fold. Left: the three arenas, named in the figures by the rotations that map their landmark layout onto itself (C_1, none: the asymmetric arena; C_2, the half-turn; C_4, the quarter-turns); shades of blue mark distinct landmarks, and the dashed outline marks one region from which the rest of the arena is generated by those rotations. Right: when the rotations leave the views and the movement input unchanged, twins give identical inputs, so the network gives them one state. Formally, its map is a map of the quotient X/G of the arena X by the group G of those rotations, and the quotient map π sends each set of twins (an orbit of G) to one state. A head-direction signal that differs between twins breaks the chain (orange). (**c**) The consequence for identical rooms. At matching spots (yellow place field, rat) the head-direction signal reads the same when the rooms are translated and differently when they are rotated, so the account predicts repeated fields in translated rooms and allows separate fields in rotated ones; green, the animal studies that report each result. Colours: blue, spatial layout; orange, heading; navy, symmetry and fold; green, animal data.

of twins out of the decoder's training (the path-held-out decoder; Methods), except where a section says otherwise.

Two one-bit signals that differ only in what a half-turn does to them. If the amount of heading information decides whether twins are separated, any one-bit signal should do worse than the full heading, and two one-bit signals should do equally well. If what matters is whether the symmetry changes the signal, two one-bit signals can give opposite maps. We built such a pair. Each network receives a 7×7 view and a five-dimensional movement vector, its speed and its heading among east, south, west and north. We replace the heading part by a fixed 4×4 transform of itself, which keeps its size and its overall activation (Table 1, Fig. 3a; Methods). *axis* groups east with west and north with south, so a half-turn leaves it unchanged. *flip* groups east with south and west with north, so a half-turn reverses it. The two differ in one more respect: every quarter-turn of the agent changes *axis*, but only half of them change *flip*, so the two signals report

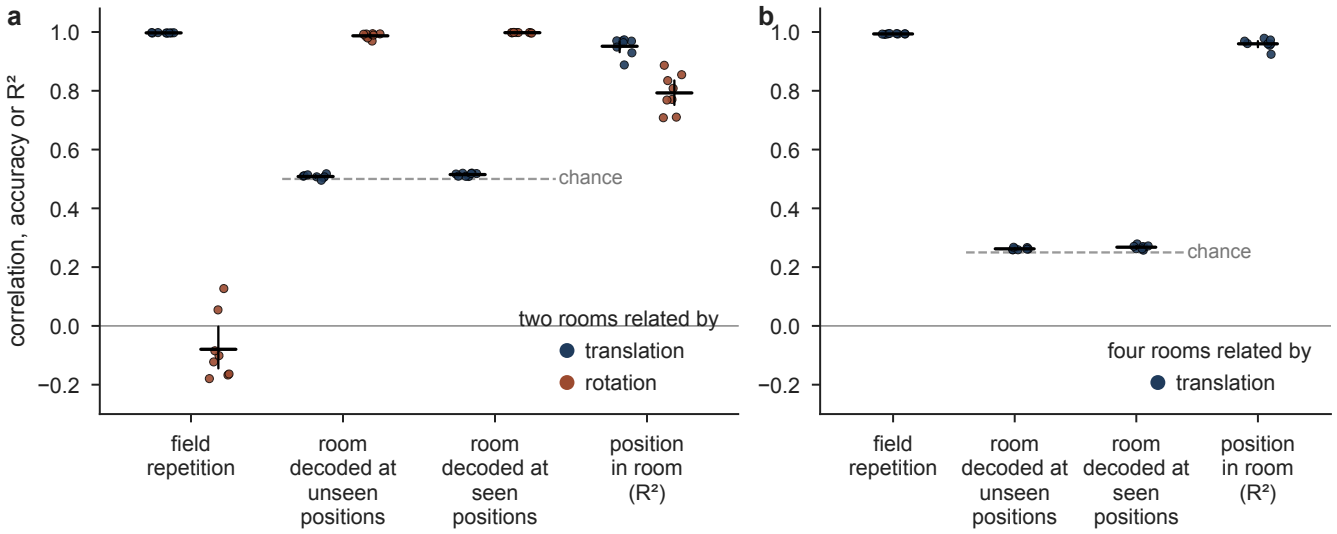


Fig. 2. Identical rooms share one map when translated and two when rotated. (**a**) The two-room maze, $n = 8$ networks per arrangement, one dot per network. Repetition score (median correlation of a unit's firing map between the two rooms), room decoding at positions the decoder was not trained on (chance 0.5), and decoding of position within a room (R^2), for translated and rotated rooms. Under translation fields repeat and the room is not decoded; under rotation fields do not repeat and the room is decoded; position within a room is kept in both. (**b**) The four-room maze of Spiers et al. (1), $n = 8$ networks: four identical rooms related by translation share one map (repetition 0.99, position within a room $R^2 = 0.96$), and the room decodes near chance (0.25, dashed). Which networks and decoder stand behind each value is listed in Methods (Table 4).

turns differently. The asymmetric arena controls for this, because there the half-turn is no symmetry and any effect of how the signals report turns would still show.

Trained on prediction alone, the networks develop units with localised firing fields, as Levenstein et al. (7) reported for this objective. In the asymmetric arena under *full*, 474 to 485 of the 500 units of a network have a place field ($n = 10$ networks; criteria in Methods).

One bit of heading separates twins or folds them, depending on what a half-turn does to it. Single units show the two signals apart before any decoding, and they also show why a decoder is needed (Fig. 3b). We measured how much a unit's rate map changes when it is turned by a half-turn, as the correlation between the map and its rotated copy, over the 40 most spatially informative units of one network per condition. In the half-turn arena, maps under *flip* change completely (median correlation -0.015). Maps under *axis* hardly change (0.98): a unit that fires at a place fires again at its twin. Under *const* in the quarter-turn arena, maps repeat at all four twins (quarter-turn correlation 0.98). The surprise is the asymmetric arena. There the landmarks tell twins apart, yet maps under *axis* are almost as symmetric (0.90). A symmetric rate map follows the signal, not the fold, so whether a map is folded has to be read from the population.

The decoder reads it, and the comparison between the two one-bit signals is the central result (Table 2, Fig. 3c, d). Both give the network one bit about its heading. Yet in the half-turn arena, networks trained with *flip* separate twins at 0.952, while networks trained with *axis* sit at chance (0.501); the two groups of ten networks do not overlap (Methods). The reason is that the *flip* bit takes different values at a position and at its twin, so it tells the network which of the two it is at, whereas the *axis* bit takes the same value at both. This is not a general weakness of *axis*. In the asymmetric arena, where the landmarks tell twins apart, the two signals perform identically (0.964 against 0.963). And one bit can do the work of two: *flip* comes within 0.015 of the full heading in the half-turn arena.

The result holds for nonlinear decoders, in a ring-shaped arena, with a heading signal corrupted on 30% of steps, and with a heading that the network must build from its own turns (Supplementary Information).

A folded map is still a map, of the folded arena. A decoder at chance is a negative result, and a code that had simply broken down would give one too. The theorem predicts something sharper. It caps a code folded by a set of $|G|$ rotations at $1/|G|$, so a code that still maps the folded arena should reach that cap, while a broken code sits at chance. The quarter-turn arena tests this (Fig. 4a). Here a decoder must name which of four twins the agent occupies, so chance is 0.25. *axis* is unchanged by the half-turn but not by the quarter-turn, so its cap is $1/2$: it can tell the two half-turn pairs apart and no more. It comes close to the cap (0.472), far above chance. *const* is unchanged by every quarter-turn, so its cap, $1/4$, equals chance, and its accuracy sits there (0.248); position within one quarter of the arena is still decodable from it (Methods). Neither value was fitted. *full* and *flip* stay high. Each signal folds the map along exactly the symmetries that leave it unchanged.

What survives the fold is a map of the folded arena. A linear readout recovers the agent's position within one half of the arena: for 104 of the 112 networks it reaches $R^2 \geq 0.81$, and the eight exceptions, *const* in the quarter-turn arena, keep position within one quarter instead (tested in four networks; Methods). Twins become one point. The cosine between the population states at a position and at its twin is 0.990 under *axis* in the half-turn arena, against 0.741 under *flip* (Fig. 4b). Of the four conditions measured, only *axis* in the half-turn

Table 1. The four head-direction signals. *axis* and *flip* carry one bit each and differ in whether a half-turn leaves them unchanged.

signal	bits	unchanged by a half-turn	unchanged by a quarter-turn
full (E, S, W, N)	2	no	no
flip ({E,S} vs {W,N})	1	no	no
axis ({E,W} vs {N,S})	1	**yes**	no
const	0	yes	**yes**

Table 2. Disambiguation accuracy: how well a decoder names which of two twins (positions a half-turn apart) the agent occupies. Chance is 0.500. $n = 10$ networks per cell for the asymmetric and half-turn arenas, $n = 8$ for the quarter-turn arena. Path-held-out decoder.

arena	*full*	*flip*	*axis*	*const*
asymmetric (C_1)	0.967	0.963	0.964	0.960
half-turn (C_2)	0.967	0.952	**0.501**	**0.502**
quarter-turn (C_4)	0.970	0.917	**0.505**	**0.506**

arena puts twins closer together than neighbouring positions (Fig. 4c–e). Under *axis* and *const* the fold is exact: a trajectory and the same trajectory turned by a half-turn drive identical states in every network (Methods).

Prediction, not reconstruction, makes the networks use the heading. The theorem says which twins no network can separate; it does not say that a network will separate the rest. The clearest demonstration is a learner that predicts well and still separates no twins. We trained a causal transformer that sees the last 64 steps, and for comparison a GRU and an LSTM, all to predict the next view in the half-turn arena (Supplementary Information). Given the full heading, the transformer predicted the next view with a lower error than both recurrent networks, yet a linear decoder could not read from its states which twin the agent occupied (0.492, $n = 5$). Predicting the next view therefore does not require knowing which of two twins the agent is at, and the transformer solves the task without it. The recurrent networks do learn it: the GRU and the LSTM separate twins under *full* (0.957 and 0.966) and under *flip*, and fold them under *axis* and *const*, as the theorem requires.

Our recurrent networks separate about as much as their current input allows, and no more. For each arena and signal we can count, with no training at all, the fraction of position–heading pairs whose current view and heading signal already differ from those of the twin. That count sets a limit for any decoder of the current input. The trained networks match it to within 0.024 on average across all twelve combinations of arena and signal (Supplementary Information). The limit is even visible where nothing is folded. Under *axis* in the asymmetric arena, 6.6% of states happen to share their input with the state a half-turn away, so the networks reach 0.964 rather than 1.000. Memory could carry a network beyond this limit: the exact ceiling from the full history of inputs is 1.000 in the asymmetric arena, and a smooth place code read through the same decoder reaches 0.999. The networks stay 0.03 to 0.05 below it, which suggests that they separate twins mostly from what they currently see and the heading they currently receive (Supplementary Information).

If the objective never requires twin identity, why do the recurrent networks learn it? Our reading, which the next result supports but does not prove, is that prediction makes them use the heading signal, and a network that uses a signal that differs between twins separates them as a by-product. We kept the network, the signal and the data and changed only the prediction horizon k, the number of future views predicted at each step. At $k = 0$ the network only reconstructs its current view, and with every one of the four signals it separates twins no better than a code that cannot separate them at all (Supplementary Information). Its inputs are not the problem: under *full* they distinguish $1{,}228$ of the $1{,}296$ position–heading pairs. Reconstructing the present needs neither the heading nor which twin you are at, and the autoencoder learns neither. With one step of prediction, *full* and *flip* separate twins and *axis* and *const* do not, and longer horizons add little. An objective that needs memory without prediction, reconstructing the view seen one or three steps earlier, formed no map at all (Methods).

Beyond the forced fold: the networks keep a knowable room for about fifteen steps. The theorem says when a network *must* fold. The identical rooms ask a second question: when the room can be known, does the network keep it? To separate the two cases we computed, for every step, the best accuracy any learner could reach from the full history of inputs, by running the exact Bayesian filter over the maze's states (Methods). In translated rooms this ceiling is 0.500 on 79% of room steps ($239{,}314$ of $301{,}357$), because the agent has not yet seen the junction of the corridor, the only place where the two rooms differ. On those steps the fold is forced. On the remaining 21% the ceiling is 1.000: an ideal observer knows the room and never forgets it.

On the knowable steps the networks remember the room, and then lose it (Fig. 5b). Room decoding stays at 0.96 or above from the fourth step after the last input that told the rooms apart (the earliest a room step can come) to the tenth, falls to half of what the ceiling allows above the floor after 14.6 steps (95% interval 14.0 to 15.2, $n = 8$ networks), and is back at the floor after about thirty, where the networks keep a fraction 0.010 of what the ceiling allows (95% interval -0.009 to 0.029). The loss costs them something: remembering the

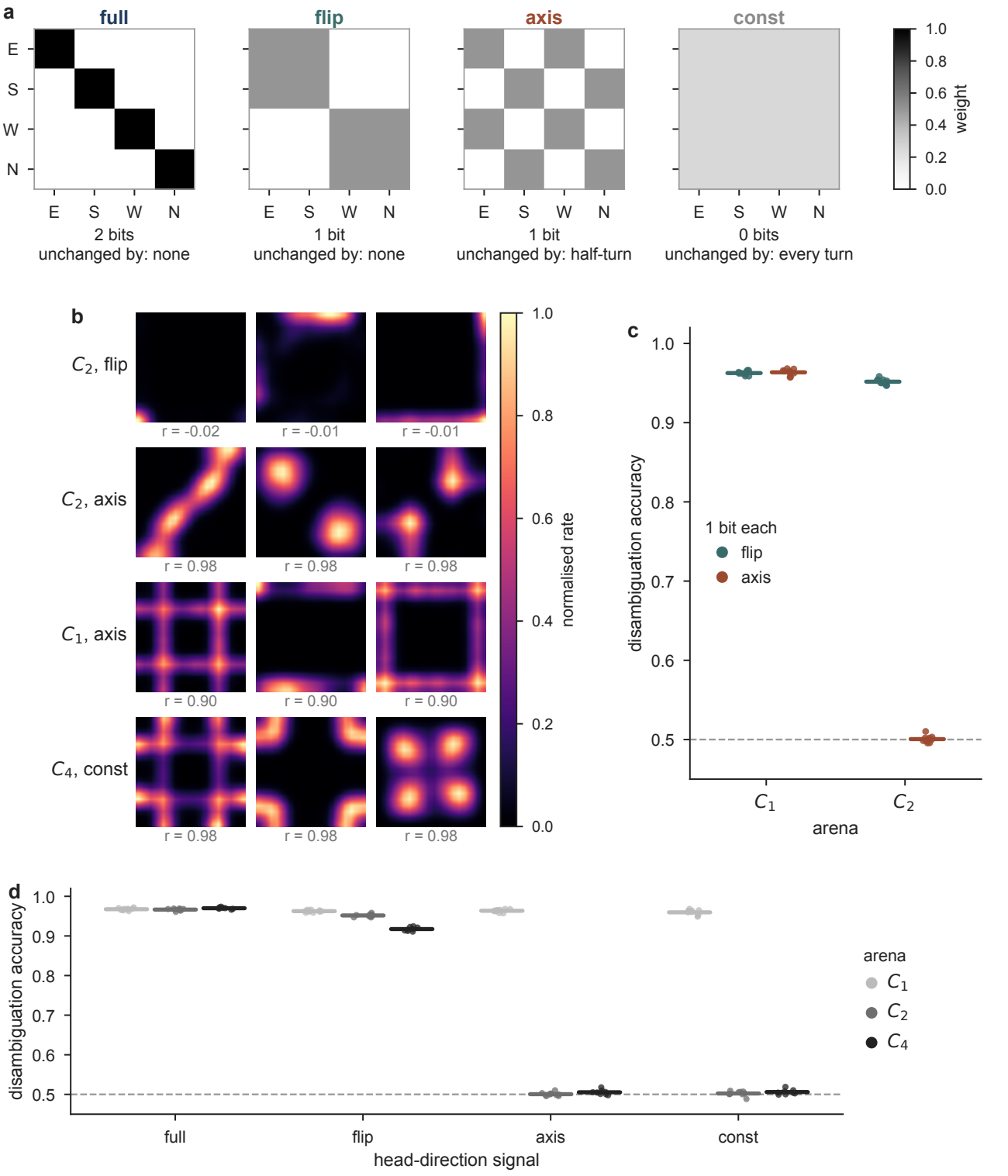


Fig. 3. One bit of heading separates twins or folds them. (**a**) The four head-direction signals, as 4×4 transforms of the heading (east, south, west, north); grey level is the weight. *flip* and *axis* reduce the heading to one bit with different groupings. (**b**) Rate maps of example units from one network per condition (peak-normalised). The units are the three place cells (Methods) whose half-turn correlation r, printed under each map, lies closest to the median of the 40 most informative units of that network, so the three values in a row are similar by construction. Under *axis* maps are nearly unchanged by a half-turn in the half-turn arena (C_2) and also in the asymmetric arena (C_1), where nothing is folded; under *flip* they are not; under *const* in the quarter-turn arena (C_4) they repeat at all four twins. (**c**) The matched pair: *axis* and *flip* are indistinguishable in the asymmetric arena and separate completely in the half-turn arena. (**d**) Disambiguation accuracy of every trained network (one dot per network; bar, group mean; chance 0.5, dashed), by signal and arena. Path-held-out decoder; $n = 10$ networks per signal in C_1 and C_2, $n = 8$ in C_4; data of Table 2.

room would lower their prediction error on these steps by up to 6.1% of its converged value (Methods). We call this *over-*

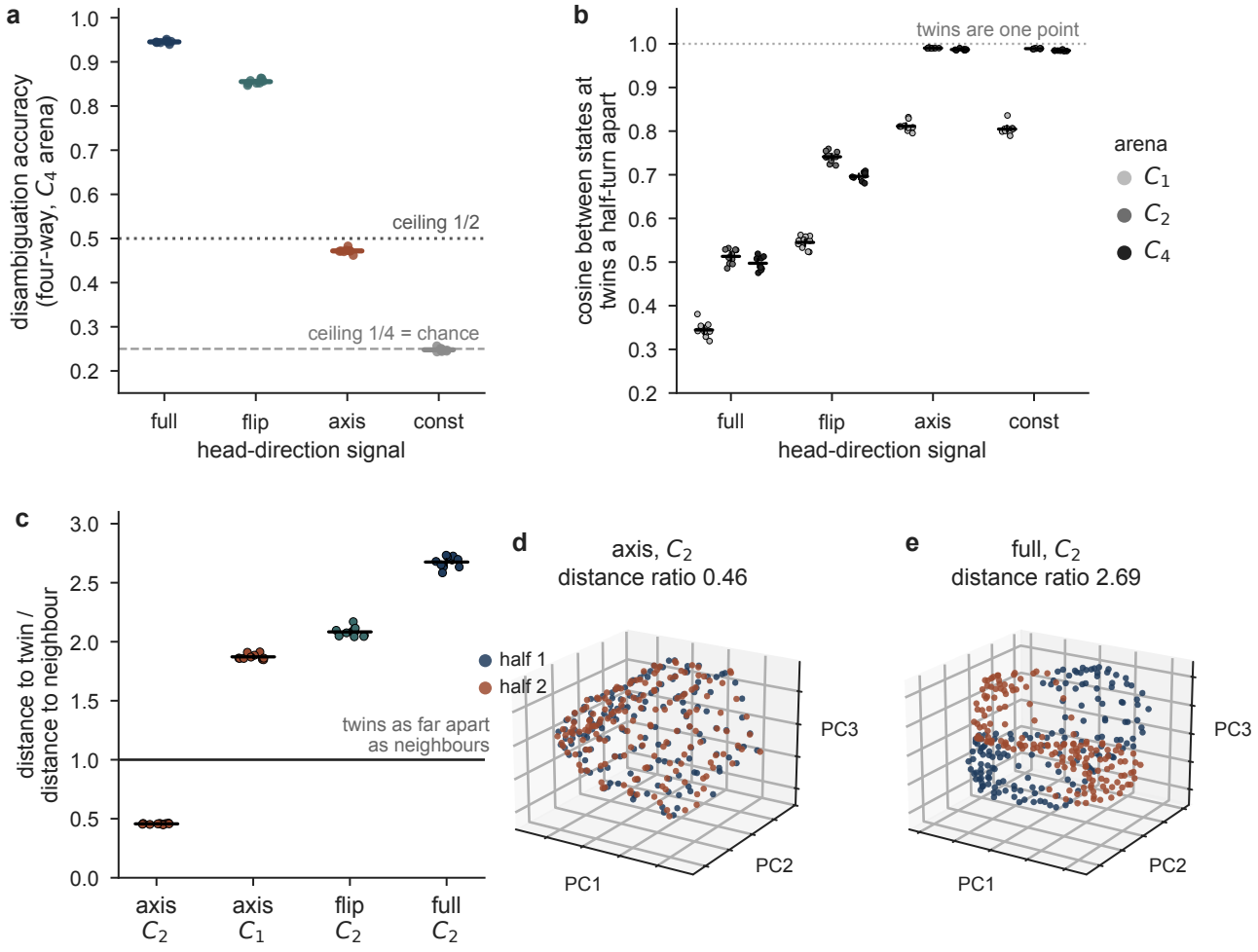


Fig. 4. A folded map is a map of the folded arena. (**a**) Four-way disambiguation accuracy in the quarter-turn arena (one dot per network; bar, mean; chance 0.25, dashed). A code that cannot tell $|G|$ twins apart is bounded by $1/|G|$: *const*, unchanged by every quarter-turn, falls to the $1/4$ ceiling, and *axis*, unchanged by the half-turn only, falls to $1/2$ (dotted lines). The ceilings are predictions, not fits. Path-held-out decoder; $n = 8$ networks per signal. (**b**) Cosine between the population states at a position x and at its twin $\rho^2 x$, one dot per network. Under *axis* in the half-turn arena the two positions are one point; under *flip*, which carries the same one bit, they are not. (**c**) Distance between the states at a position and at its twin, in units of the distance to an adjacent position. Below one, twins are closer than spatial neighbours; only *axis* in the half-turn arena is below one. One dot per network, $n = 10$ per condition. (**d**, **e**) Population states of the most and the least folded condition, one point per position, projected on three principal components and coloured by half of the arena: the two halves lie on top of each other under *axis* and apart under *full*. Panels (b)–(e) use the original sweep (Methods, Table 4).

folding: a code that merges states its inputs have already told apart.

The memory has a place in the code, and removing it removes the memory and nothing else (Fig. 5c). The difference between the states of a path and of the matched path through the other room occupies 20 to 24 dimensions of the 500 (95% of its variance). Projecting these dimensions out at every step drops room decoding to the floor (0.46 to 0.52, against a floor of 0.47 to 0.53, 8 networks), while position within a room is still decoded at $R^2 = 0.997$. Projecting out a random subspace of the same size and variance leaves the room readout within 0.014 of intact.

Over-folding is not forced by the network's capacity or by the length of its training window (Fig. 5d). We retrained the networks with one change at a time (6 seeds per arm, each against the same seeds trained unchanged). With an added loss that asks the network to report the room on every step, it keeps the room indefinitely: beyond thirty steps it keeps a fraction 0.990 of what the ceiling allows (95% interval 0.979 to 1.000), against 0.021 for the same seeds trained unchanged, higher in all six pairs, and its map is intact ($R^2 = 0.995$). A network of this size can hold the memory. With training paths and backpropagation three times as long, the memory is no longer: its half-life is 15.0 steps (interval 14.4 to 15.6) against 14.8 (interval 14.1 to 15.7) for the same seeds (paired Wilcoxon $p = 0.69$). The training window does not cut it short. Our reading, which these controls support but do not prove, is that the prediction objective rewards the memory too rarely and too weakly: the room changes the next five views on only 0.8% of the steps where it is known.

Which steps the networks keep the room on supports this reading. If the networks kept what prediction needs, they would keep the room where it changes the upcoming views and drop it elsewhere. They do not. Six to ten steps after the junction they keep the room where it changes nothing in the next five views (0.981), and more than twenty steps after

it they lose the room where it does matter (0.538, against a floor of 0.534). Retention follows the time since the informative input, not its usefulness for prediction (Supplementary Information).

Silencing the heading folds the map only where a symmetry exists. Go back to waking up in two identical rooms, and now take the compass away. If the rooms are shifted copies, you lose nothing you needed, because the compass never told them apart. If one room is turned, you lose the one thing that did. And in a room with distinct landmarks you lose some sense of direction but never doubt where you are. A head-direction lesion likewise removes two things at once, the information about heading and the difference the signal makes between twins, and the account predicts different consequences for each. Losing the information should degrade the map in every arena. Losing the difference should fold the map, but only where the arena has a symmetry.

We silenced the head-direction signal of networks trained with the full heading, by setting it to zero on a fraction of steps at test time. We silenced it rather than randomising it because an ablated head-direction system carries no signal (9). This is an acute silencing, not a lesion: the networks never saw a zero heading in training, so part of any loss may reflect the unfamiliar input, and we therefore compare arenas, which share that property. The two consequences separate (Fig. 6). Silencing shrinks the map of the folded arena, the share of the code explained by which set of twins the agent is in, by 31% to 38% in every arena, and the loss is not ordered by symmetry (Methods). Disambiguation accuracy behaves differently. Read here with the orbit-held-out decoder, which reads about 0.55 on a fully folded code and can partly recognise paths (Methods), it falls to that level in the half-turn and quarter-turn arenas (0.566 and 0.567) when the signal is silenced on every step, and holds at 0.862, an upper estimate, in the asymmetric arena, where every network stays above every network of the symmetric arenas. The number of fields per unit splits the same way: it falls by 11% in the asymmetric arena and rises by 16% and 20% in the symmetric ones. So silencing degrades the map about as much in an arena with no symmetry as in one with a fourfold symmetry, and it folds the map in only one of them.

The two-room maze shows the same interaction between lesion and arrangement that rats show. In a second set of networks (six per maze; Methods), with the signal silenced on every step, fields repeat across rotated rooms (repetition 0.958, against −0.137 intact), and the room can no longer be decoded on held-out paths (0.502, against 0.996), although position within a room is still read out (Methods). In translated rooms the same silencing changes little (repetition 0.982, against 0.997), because there the signal never separated the rooms. In the original two-room networks, silencing on a fraction of steps gives a graded version of the same pattern (Fig. 6c): with the signal silenced on up to 0.6 of steps, repetition across translated rooms stays at 0.996, and across rotated rooms it rises from −0.198 to −0.025. The Discussion compares these results with the lesion experiments.

A fold decides what can be learned from the map. If you cannot tell two rooms apart, you also cannot learn that the food is in only one of them: whatever you learn in one room, you learn in both. A downstream area that learns from a folded map is in the same position, because two positions that share a state get the same output from any readout of that state. We tested this with linear readouts trained on frozen networks, holding whole trajectories out of each readout's training (Methods).

A goal taught at one place spreads to exactly the twins the head-direction signal cannot tell apart (Fig. 7a). We trained a readout on a goal placed in one quarter of the quarter-turn arena and measured the value it gives each of the goal's three twins, relative to the goal itself (1: carried in full; 0: kept in place). Under *const* the goal reaches all three twins (0.976 to 1.011). Under *axis* it reaches the twin a half-turn away (1.007) and neither twin a quarter-turn away (−0.000 and −0.004). Under *full* it stays where it was taught (0.018 or less). The theorem requires the goal to reach the twins the signal cannot tell apart. That it goes no further is not required by the theorem; we measured it. Under *flip*, which a quarter-turn leaves unchanged for half of the headings, some value reaches the quarter-turn twins, but unequally (0.209 and 0.083), which the account does not explain (Supplementary Information).

The same rule decides what cannot be learned. In the two-room maze, a temporal-difference (TD) readout learned a value function on the frozen code, with reward in a goal zone of room A, and we asked whether its value tells that zone from the matching zone of room B (Fig. 7d). Under rotation it does, and every network reaches criterion within 1 to 8 episodes. Under translation it never does: after 1,000 episodes the area under the ROC curve is 0.499, and the goal's value is carried into room B in full (1.001). The readout is not at fault, because it learns position within a room under both arrangements (Methods), and the value it carries into room B is what the fixed point of TD learning on these codes predicts (0.999).

Sharing a state also has a cost. Under translation, value learned in room B lands on the states that room A shares with it, so the readout's discrimination within room A first peaks and then falls, by 0.023 on average against 0.002 under rotation (Fig. 7e). Blocking the TD updates made in room B shrinks the fall to 0.003, and the fall is smaller in every translation network, so experience in the twin room is the cause (Methods). Silencing the head-direction signal has the matching effect on learning (Fig. 7f). Under rotation, room discrimination falls from 1.000 to 0.521 when the signal is silenced on every step. Silencing half of the steps leaves seven of eight networks near 1.0, and one falls to 0.52 after first reaching criterion. Silencing also lowers discrimination within a room, from 0.997 to 0.906, so part of its effect is not specific to the room.

A fold can also speed learning, for goals that respect the symmetry. Kernel regression theory predicts the learning curve of a linear readout from two properties of the code: the eigenvalues of its kernel, and how the target is spread over the kernel's

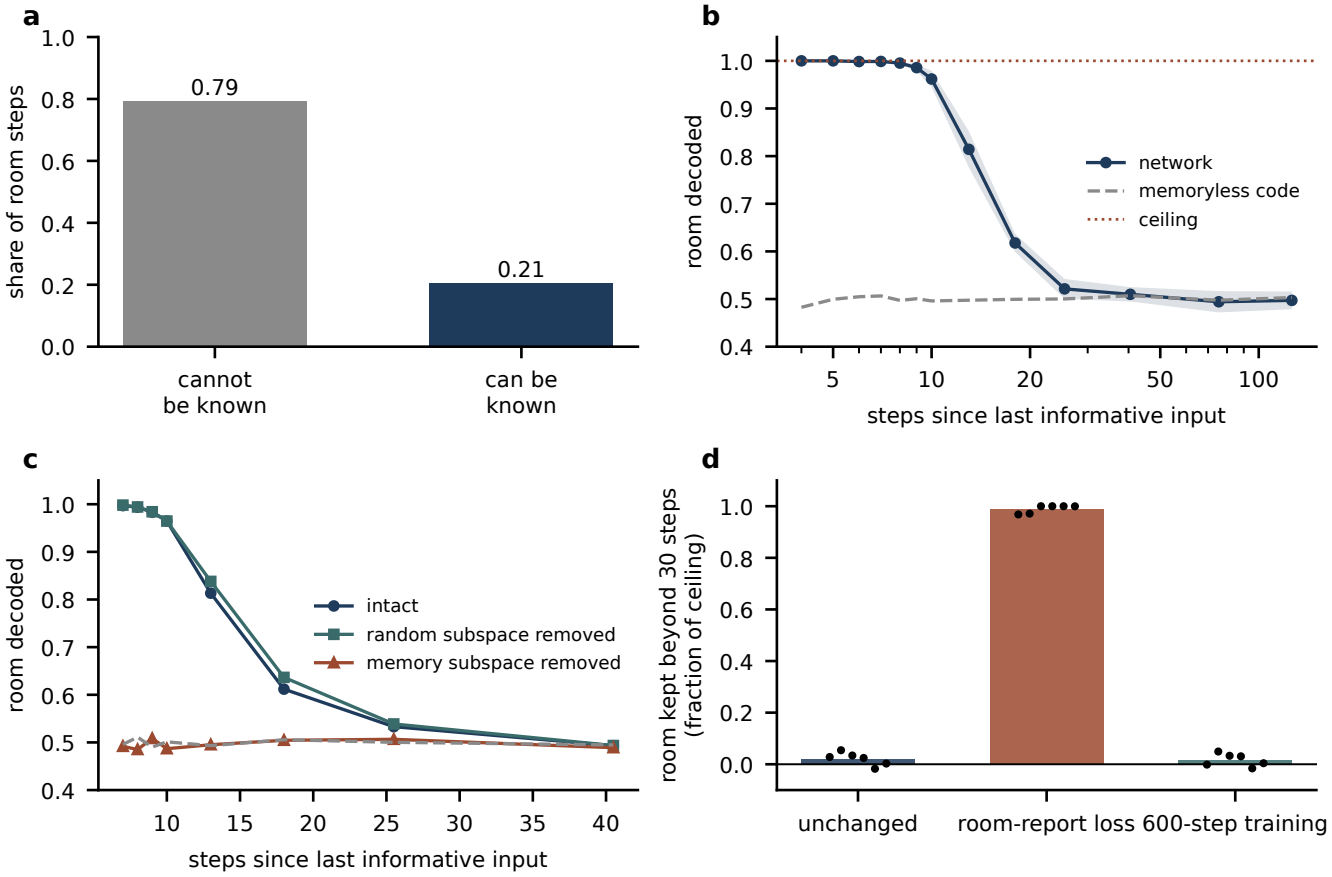


Fig. 5. Beyond the forced fold. Translated rooms. (**a**) Room steps on which an ideal observer, given the full history, cannot know the room (the agent has not yet seen the corridor junction; ceiling 0.5) and can know it (ceiling 1.0); exact Bayesian filter over $301{,}357$ room steps. (**b**) Room decoding against steps since the last input that differed between the rooms (path-held-out decoder; mean $\pm$ s.d. over $n = 8$ networks); dashed, a memoryless code through the same pipeline; dotted, the ceiling. (**c**) The same readout with the room-difference subspace projected out at every step, and with a random subspace of equal dimension and variance projected out; position within a room stays at $R^2 = 0.997$. (**d**) Retrained controls, one change each, 6 seeds per arm (dots) against the same seeds trained unchanged: the fraction of the ceiling's room information kept more than thirty steps after the last informative input, with an added room-report loss and with training paths and backpropagation of 600 steps.

eigenvectors (10). Here it predicts the measured curves with nothing fitted (the ridge is fixed in advance): in 591 of 592 learning curves the predicted curve is closer to the measured one than the curves from all of 20 shuffles of the eigenvalues (Fig. 7b, c; Methods). For a goal that is the same in both halves of the half-turn arena, a readout of the folded *axis* code reaches a relative error of 0.1 from 9.7 training positions, against 19.9 for the *full* code (predicted: 9.8 and 21.6), because such a goal already lies in the directions that the folded code keeps. A fold alone does not guarantee this: *const*, also folded, needs 29.6 positions, and the theory places all four signals in the measured order by how much of the goal lies along the leading eigenvectors of each code (Supplementary Information). For a goal placed in one half only, the folded readouts never reach a relative error of 0.1 (Fig. 7c), because the part of such a goal that differs between twins has no counterpart in a folded code.

The fold shows in behaviour too. Steered toward a goal by a readout of its own frozen code, with the view blanked on five of every six steps, an agent with the *full* code ended at the goal in 0.904 of trials, against 0.50 for an untrained network (a readout of the true position reaches 0.97, random actions 0.025). An agent with the *axis* code ended at the goal or at its half-turn twin in 0.952 of trials and at a quarter-turn twin in 0.001, so it makes exactly the errors its signal cannot rule out (Methods).

Single cells can look symmetric where nothing is folded. Everything so far rests on decoders run over whole populations, while a recording laboratory often has single cells. The example maps of Fig. 3b already warned that a symmetric map need not mean a fold, and the population of units says the same. The symmetry index of a unit is the correlation of its rate map with the same map turned by a half-turn. Under *axis* in the asymmetric arena, where nothing can fold, it averages 0.61 over all units of six networks (Fig. 8a; smoothed maps of all units, so the value differs from the 0.90 of the example network in Fig. 3b), yet twins there are decoded at 0.964. Where the map is folded the index reaches about 0.97, so it is not uninformative, but it is confounded: an unchanged heading signal raises it with or without a fold. A fold has to be tested in the population. Where a fold does occur, it is not the property of a class of cells: under *axis* in the half-turn arena, 93.1% of all units score above 0.8 (Methods).

The account also predicts what repeated fields should look like in a recording. In a city-block maze whose alleys run

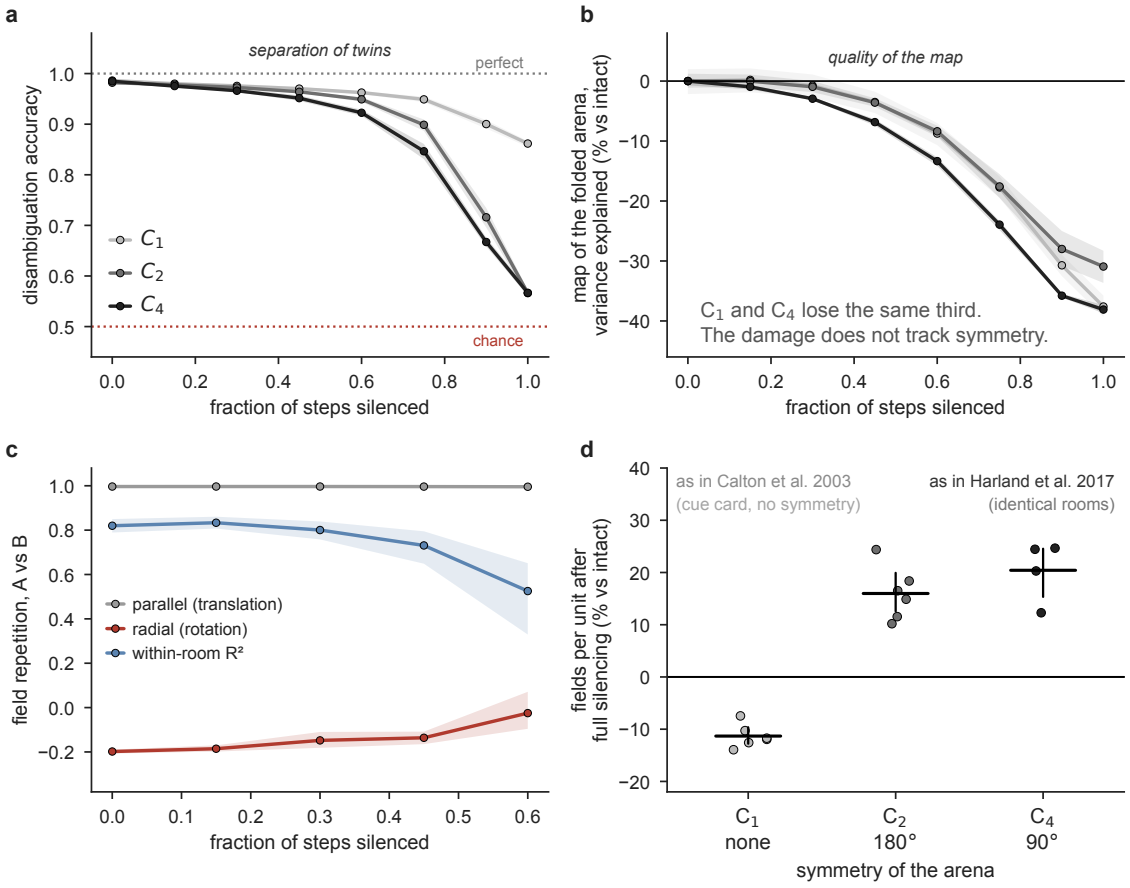


Fig. 6. Silencing the head-direction signal degrades the map everywhere and folds it only where a symmetry exists. Networks trained with an intact signal, which is then set to zero at test time on a fraction of steps. $n = 6$ networks for the asymmetric (C_1) and half-turn (C_2) arenas, $n = 4$ for the quarter-turn arena (C_4); shading, 95% bootstrap interval over networks. (**a**) Disambiguation accuracy falls to chance in the symmetric arenas and holds at 0.862 in the asymmetric arena. This panel uses the orbit-held-out decoder (Methods), whose reading on a fully folded code is near 0.55. (**b**) The variance of the code between sets of twins, that is, the map of the folded arena, falls by about a third in every arena, and the falls are not ordered by symmetry. Panels (a) and (b) together are the argument. (**c**) The two-room maze under partial silencing. In translated rooms the signal never distinguished the rooms, so repetition does not move ($0.996 \to 0.996$); in rotated rooms it rises. Position within a room (R^2, blue) stays positive, so the rising repetition is not two destroyed maps correlating with each other. (**d**) Fields per unit after full silencing, one dot per network: they fall by 11% in the asymmetric arena and rise by 16% and 20% in the symmetric arenas.

in two orthogonal orientations, alleys of the same orientation are related by a translation, and alleys of different orientations by a quarter-turn. A heading signal reads the same in two alleys of the same orientation and differently in alleys at right angles, so the fields of a repeating cell should share a preferred running direction across alleys of the same orientation but not across orthogonal ones. Networks trained on this maze show both halves of the prediction. The directional preferences of a unit's fields in alleys of the same orientation are correlated ($r = 0.62$ on average over 8 networks; Fig. 8b), and those in orthogonal alleys are not ($r = 0.02$), with the same-orientation correlation the higher of the two in every network (Methods). This pattern is not unique to a fold, however: cells that each prefer one running direction, with no fold at all, would show it too.

In the CA1 recordings of Hockeimer et al. (11) from such a maze, the same-orientation correlation is 0.23 (127 pairs of fields from 40 cells in 5 rats), a value that the directional tuning of single cells alone reproduces (Methods). The data are consistent with the account but do not test it, so this reanalysis is exploratory (Supplementary Information).

Discussion

Why does a head-direction signal keep rotated rooms apart? In our networks, not because of how much it says about heading, but because a rotation changes what it reads at matching spots. In networks that learn only to predict their next view, a one-bit signal that a half-turn flips separates twins nearly as well as the full heading, and a one-bit signal that a half-turn leaves unchanged does not separate them at all. Where the inputs cannot tell twins apart, the map folds. The fold is not a breakdown: the folded code is an accurate map of the folded arena, it decides which goals a readout can learn, and for some goals it makes learning faster. Nothing in the argument is specific to head direction. Any self-motion signal separates only the twins at which it reads differently, and speed, which no rotation or translation changes, separates none of them. Head direction is simply the signal we could vary cleanly.

The animal record, read as a story about symmetry. Read with this account in mind, the animal experiments tell a consistent story. Rats in a row of identical compartments,

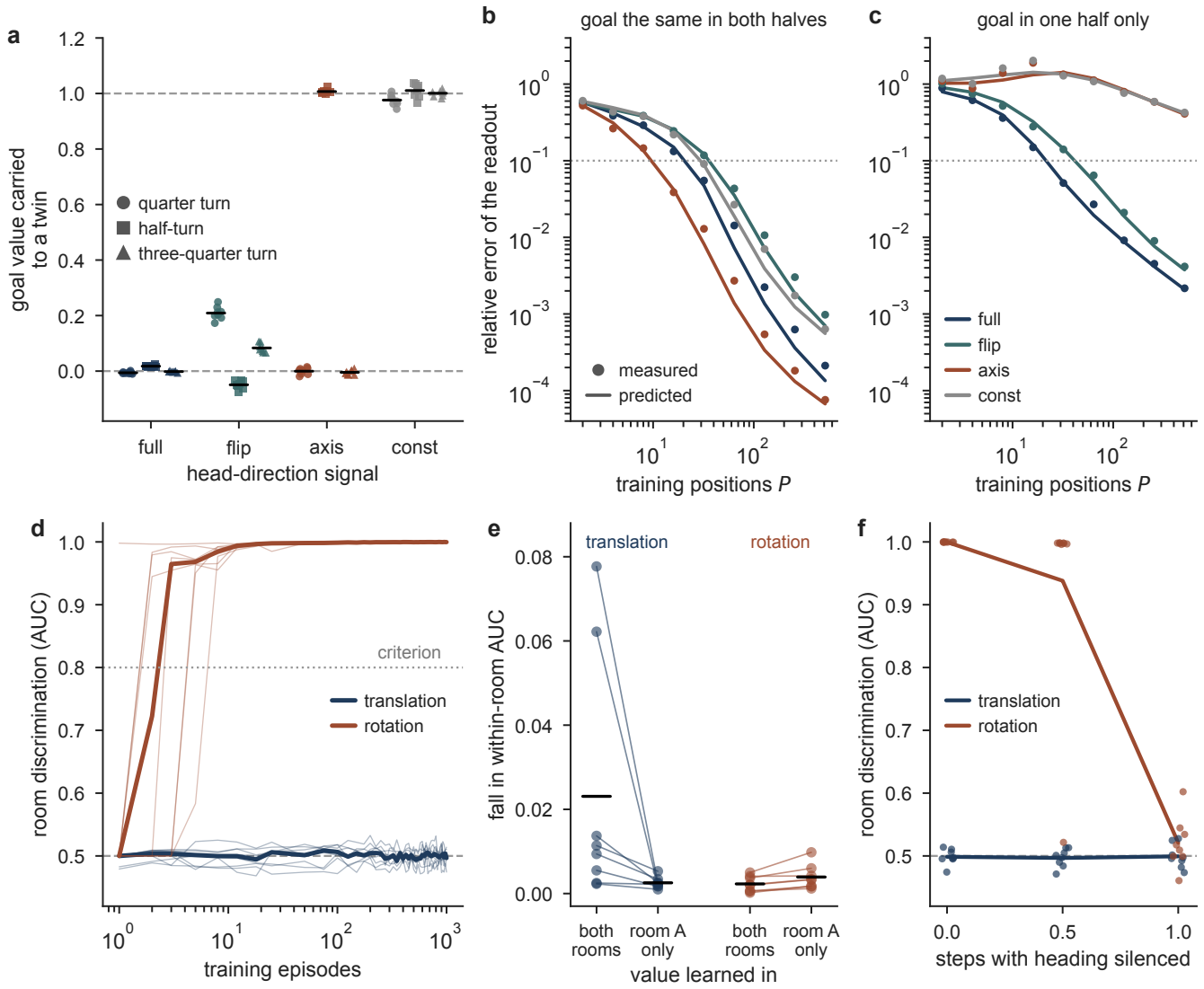


Fig. 7. A fold decides what a readout of the map can learn. (**a**) Goal value carried from one quarter of the quarter-turn arena to each of the goal's three twins by a ridge readout (1: carried in full; 0: kept in place; one marker per network, $n = 8$ per signal; bar, mean). The value reaches exactly the twins that the head-direction signal cannot tell apart: none under *full*, the half-turn twin under *axis*, all three under *const*. (**b**, **c**) Learning curves of a ridge readout in the half-turn arena, for a goal that is the same in both halves (b) and for a goal placed in one half only (c): relative error against the number of training positions. Dots, measured (mean of $n = 10$ networks); lines, predicted from the eigenvalues of each code with nothing fitted; dotted line, relative error 0.1. (**d**) A TD value readout in the two-room maze: discrimination of the rewarded zone of room A from the matching zone of room B (area under the ROC curve) against training episodes. Thin lines, single networks ($n = 8$ per arrangement); thick lines, median; dashed, chance; dotted, criterion (0.8). (**e**) Interference: fall of the within-room discrimination from its peak to its final value, when value is learned in both rooms and when the updates made in room B are blocked. Lines join the two conditions of one network. (**f**) Room discrimination by the value readout after the head-direction signal of the trained network is silenced on a fraction of steps (dots, single networks; lines, means).

each a shifted copy of the next, repeat their place fields from one compartment to the next (1). Turn the compartments to face different ways and the repetition largely disappears, even after the orienting landmark is removed, and rats learn to tell the compartments apart more readily than when they lie side by side (2). The same contrast appears with just two identical boxes: side by side, their maps do not separate, and rotated by a half-turn, they separate completely (12). Our networks fold translated rooms and separate rotated ones, and their value readout never learns to tell translated rooms apart, a more extreme form of the rats' slow learning.

The lesion experiments test a signed prediction of the account. Where the signal never distinguished the rooms, removing it should change nothing; where it was what distinguished them, removing it should fold them. Harland et al. (3) lesioned the lateral mammillary nuclei, which abolishes the head-direction signal in the anterior thalamus (9, 13), and found this pattern: repetition across parallel (translated) compartments was unchanged, and across radial (rotated) compartments it came back, and it tended to come back more in animals with larger lesions. Their radial maps were rotated into alignment before being compared, in an apparatus with no polarising cue, so their measure asks precisely whether the code is unchanged by the rotation that relates the compartments. Acute silencing of the signal in our networks reproduces the direction of both results, with nothing fitted to these data, although we built the model after they were published. Behaviour follows the same direction: fewer le-

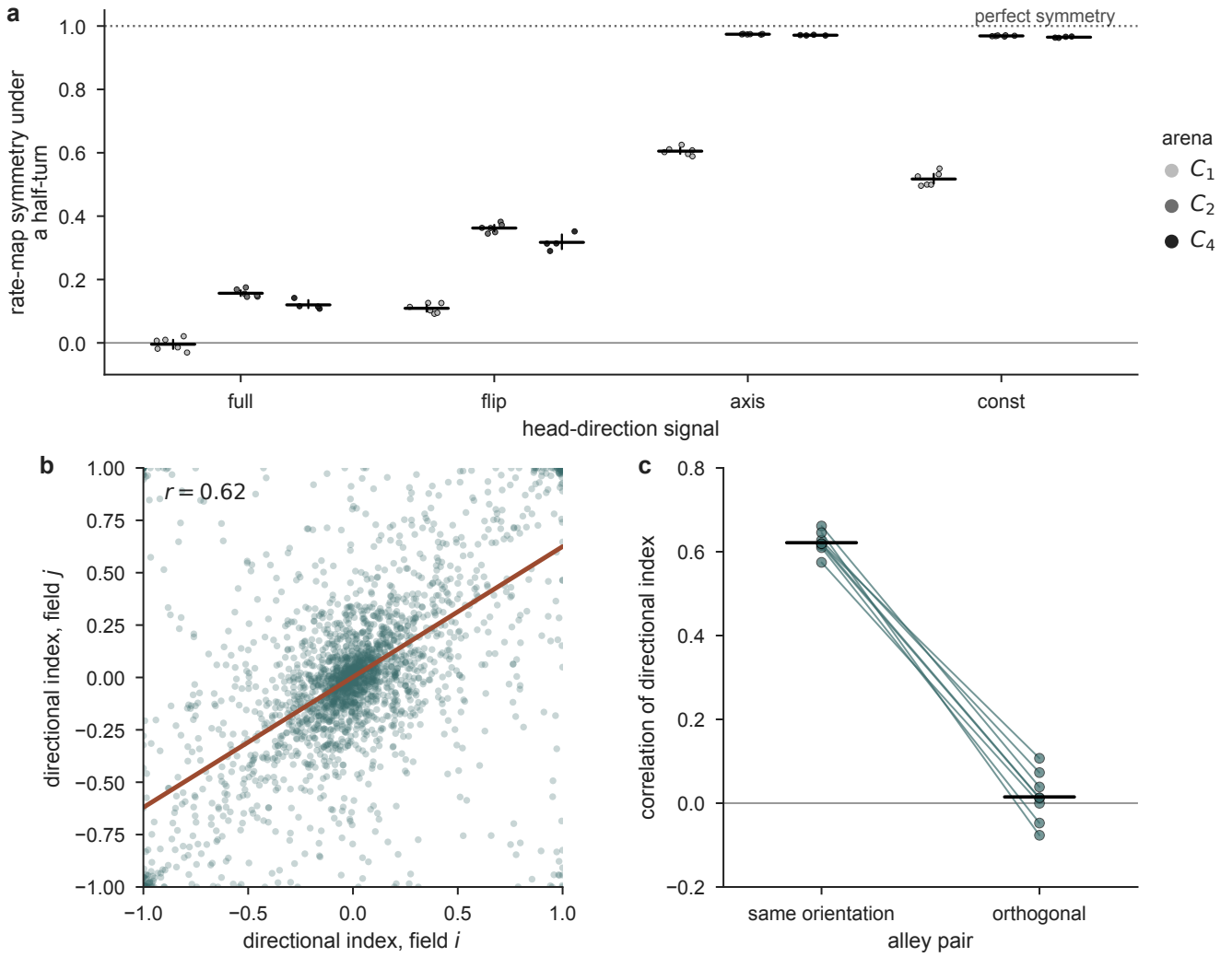


Fig. 8. What a recording can and cannot show about a fold. (**a**) Symmetry index of single units (correlation of a unit's rate map with its own half-turn rotation), by signal and arena; bars, mean ± s.e.m. over networks ($n = 6$ per signal for the asymmetric (C_1) and half-turn (C_2) arenas, $n = 4$ for the quarter-turn arena (C_4)). An unchanging signal alone drives the index to 0.61 in the asymmetric arena, where nothing can fold, so the index describes cells but does not test for a fold. (**b**) The directional tuning of repeated fields, in networks trained on a city-block maze: the directional index of one field of a unit against that of another of its fields in an alley of the same orientation ($r = 0.62$; 8 networks, $3{,}985$ units; scatter subsampled for display; line, least-squares fit). (**c**) Per network, the correlation for fields in alleys of the same orientation against that for fields in orthogonal alleys; lines join one network.

sioned rats learned to discriminate rotated compartments (4), and our value readout fails in the same arrangement. The matching experiment on translated compartments, where the account predicts that a lesion leaves learning unchanged, has not been done.

The same logic reconciles two lesion studies that seem to disagree. Lesions of the anterodorsal thalamus and postsubiculum in a cylinder degraded place fields but did not significantly increase the number of fields per cell (14). That cylinder held a cue card at a fixed bearing, which breaks its rotational symmetry. A lesion cannot create a symmetry: where another cue already tells positions apart, removing the head-direction signal degrades the map and folds nothing, and only where the signal was the sole distinction does the lesion fold the map. Our networks show the increase in the symmetric arenas (16% and 20%; Fig. 6d). In the asymmetric arena the field count falls by 11%, where Calton and colleagues saw no change.

An experiment that can be run now. The account predicts different outcomes in one apparatus, and the apparatus already exists. LaChance and Hasselmo (15) placed two identical cue cards on opposite walls of a square, and head-direction cells in postrhinal and retrosplenial cortex began to fire in two opposite directions, a cortical signal that a half-turn leaves unchanged, much like *axis*. It is tempting to predict that place fields then repeat across the two halves of the square. For an intact rat, the account predicts that they do not. In the same animals, head-direction cells in medial entorhinal cortex and parasubiculum did not detectably become bidirectional, so the classical signal, which tells the two halves apart, is still there, and the lesion evidence above suggests that it is the signal the map reads. An intact rat in the two-cue square should therefore anchor its map to one of the two orientations the room allows, as disoriented mice do in a symmetric chamber (16), rather than fold it.

Remove the classical signal, and the prediction changes.

Crossing one cue card or two with a sham or a head-direction lesion gives four conditions, and the account predicts repeated fields in exactly one of them, two cues with a lesion, with nothing free to fit (Fig. 9). Two of the four have close precedents: the intact single-cue arena gives an ordinary place map, and the lesioned single-cue arena is the cylinder of Calton et al. (14). Both two-cue conditions are open for place cells. Repetition with a lesion and none without would support the account; repetition in both, or in neither, would count against it. The prediction for the lesion does not depend on the symmetric cortical cells, because a signal that is symmetric and a signal that is absent both fold the map. The prediction for the intact rat does depend on one assumption, that the entorhinal signal stays unidirectional after a single-cue baseline. The cortical effect also shows hysteresis (15), so the conditions should be counterbalanced rather than run in sequence.

Cortex has a head-direction signal that rotation leaves unchanged. The symmetric cortical signal deserves its own word, because it answers an obvious objection: nothing in the brain looks like *axis*. Something does. Retrosplenial cells whose tuning follows the symmetry of the layout, bidirectional in a twofold environment and fourfold in a fourfold one, appear with experience (17), and duplicating a cue card produces the same bidirectional tuning in postrhinal and retrosplenial cortex (15). These cells coexist with classical head-direction cells that keep a single preferred direction, and, as Zhang et al. (17) point out, that surviving global signal allows the two halves to be told apart. The account therefore does not say that the head-direction system folds the map; it says that a predictive code folds along every symmetry that leaves the signal *it reads* unchanged, and the lesion results suggest that the hippocampus reads the classical signal. Our networks add a caution for anyone reading tuning curves. Under *axis* in the asymmetric arena, 93% of the directional tuning of an average unit lies in its two-peaked component (Methods), yet twins stay separated, so a symmetric tuning curve says nothing on its own about the map. That suggests one resolution of the apparent paradox in LaChance and Hasselmo (15), where postrhinal cells became bidirectional while the firing of entorhinal and parasubicular grid cells did not change.

Relation to other models of the spatial code. Parallel and radial compartments are built from the same parts and are locally identical, so the contrast between them discriminates among models (Table 3). The closest alternative is the boundary-vector account of repetition (18). Our networks do grow boundary-vector cells, as predictive networks do (19), but a boundary-vector cell is tuned to a wall at an allocentric bearing, and an allocentric bearing needs a head-direction signal (20). Removing the signal from our networks removes most of their boundary cells, and *axis* and *flip* support comparable boundary populations with opposite maps (Supplementary Information), so a boundary-vector account inherits the question this paper answers. The successor representation (21) is predictive too, but its states are given rather than inferred, so whether two identical compartments are one state or two is an assumption of that model, and our result says when the answer is one. In our networks, a smooth spatial kernel matched the learned codes better than either the full or the folded successor representation (Supplementary Information). The clone-structured cognitive graph gives rotated rooms unique fields when it receives an external head-direction input (22), which agrees with our result, and the Tolman–Eichenbaum machine (23) would fold whenever its action code is unchanged by the symmetry. In the language of equivariant learning, the theorem is an instance of Curie's principle: an output can break a symmetry only if an input breaks it (24). Jeffery (5) described repetition between aligned compartments as a translational symmetry that echoes the symmetry of the environment, and named directional information as the prime cue that breaks it. The matched pair sharpens that intuition: what matters is whether the symmetry changes the directional signal, not how much direction it carries.

Limitations. The model is deliberately simple, and its departures from the animal record are real. The arenas are grids with four headings, the network has no spikes, theta rhythm or separate excitatory and inhibitory cells, and the main results come from one architecture and size. Silencing is acute, at test time, and is not a model of a lesion with recovery. Each network is trained once, in a fixed arena, so it describes the code a predictive objective converges to, not the course of learning in an animal. The head-direction signal is supplied rather than learned, except in one control in which the network builds it from its own turns. The room mazes satisfy the theorem only approximately, because the corridor between rooms breaks the symmetry. The networks also fold translated rooms more completely than rats do: they keep the room for about fifteen steps after the last input that told the rooms apart ("Beyond the forced fold"), where an animal can carry its last entryway for at least five seconds (29), they show none of the small rate differences between repeated fields that Spiers et al. (1) reported, and they do not anchor to one orientation as disoriented mice do (16). A comparison with fifteen observations from the animal literature finds five reproduced (two only in direction), three partly reproduced or consistent, and seven departures (Supplementary Information). The one test on real neurons, the CA1 reanalysis, is exploratory.

Outlook. Because the map is learned, every factor that shapes it in an animal, and is confounded there, becomes a variable that can be set on its own: the layout, what the self-motion signal carries and what a symmetry does to it, the way the agent moves, and what it is trained to do. The networks here were trained once, in one arena, and training is not development. The natural next step is to raise them as rearing experiments raise animals, in worlds whose symmetries change as they grow, and to ask when a fold forms, and whether experience can undo it.

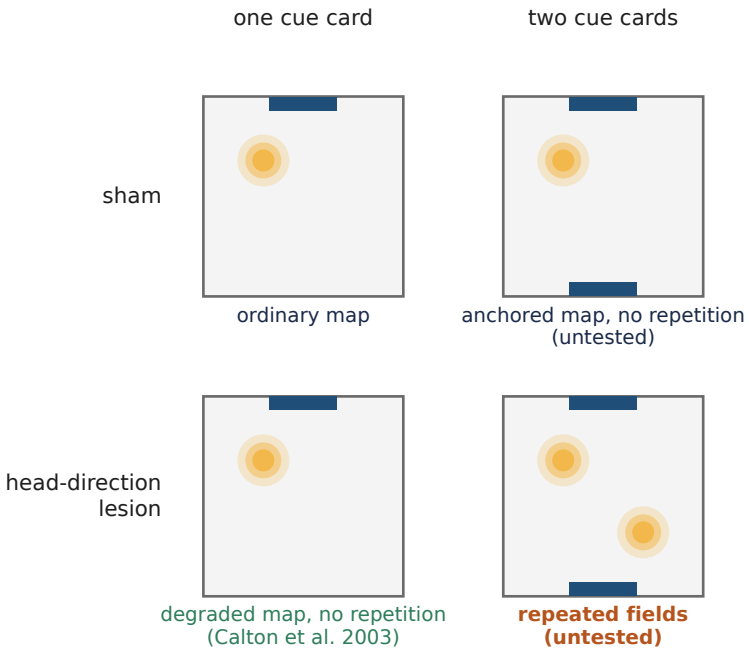


Fig. 9. A two-by-two experiment that the account predicts. A square arena with one cue card (left column) or two identical cue cards on opposite walls (right column), in sham-operated or head-direction-lesioned rats (rows). The account predicts repeated place fields across the two halves of the square (yellow fields and their twins) only with two cues and a lesion: one cue breaks the symmetry, and an intact classical head-direction signal tells the two halves apart. The intact single-cue arena and the lesioned single-cue arena (14) have close precedents; neither two-cue condition has been tested for place cells.

Table 3. Models of the spatial code and the contrast between parallel and radial compartments. The two arrangements are locally identical, so a model that reads firing from local geometry or the current view predicts the same repetition in both.

model	repetition in identical rooms	parallel-versus-radial contrast
Continuous attractor / path integration (25, 26)	not addressed	no
Boundary-vector cells (18, 27)	yes	only through a head-direction signal
Successor representation (21)	set by the given states	not addressed
Clone-structured graph (22, 28)	yes	with an external head-direction input
Tolman–Eichenbaum machine (23)	yes, if the action code is symmetric	consistent
Autoencoder (this work, control)	yes	no
Predictive recurrent network (7), this work	yes	**yes**, set by what the symmetry does to the heading signal

Methods

Two sweeps of networks stand behind most results: an original sweep and a replicate of it trained from new seeds with the same design ($n = 10$ networks per signal in the asymmetric and half-turn arenas, $n = 8$ in the quarter-turn arena; 112 networks each). Table 4 lists, for every figure and table, which networks and which decoder or readout produced it. In these Methods we use the standard terms for the objects the main text describes in words: the set of rotations that maps the arena onto itself is the group G (C_1, C_2, C_4 for the asymmetric, half-turn and quarter-turn arenas), a set of twins is an orbit of G, a region containing exactly one member of each orbit is a fundamental domain, and the folded arena is the quotient X/G. We write ρ for the quarter-turn, so that $\rho^2 x$ is the half-turn twin of x.

Architecture and training. We use the $\mathrm{pRNN}_{\mathrm{th}}$ network (7) unchanged: a layer-normalised recurrent cell, hidden size 500, truncation $T = 200$, rollout horizon $k = 5$ unless stated, observation size 147 ($7 \times 7 \times 3$), action size 5. Training uses RMSprop ($\alpha = 0.95$, $\epsilon = 10^{-6}$) with four parameter groups at per-group learning rates $\lambda\sqrt{1/H}$ for $W_{\mathrm{rec}}, W_{\mathrm{out}}$, $\lambda\sqrt{1/(n_{\mathrm{obs}} + n_{\mathrm{act}})}$ for W_{in} and 0.1λ for the bias, with bias-only weight decay; $\lambda = 2 \times 10^{-3}$, batch size 8, gradient clipped per model to norm 1, 80,000 steps.

The theorem, formally. Let a finite group G act freely on the arena's positions X, with a G-invariant start distribution, a G-equivariant observation likelihood $p(o \mid x)$, and a policy measurable with respect to the agent's belief. If the head-direction signal ϕ is G-invariant, $\phi(h + g) = \phi(h)$ for every $g \in G$, then the posterior over position given any history of observations and movement signals is G-invariant, and any decoder of the member of an orbit has accuracy at most $1/|G|$.

Proof. The action of $g \in G$ on trajectories, $\gamma \mapsto g \cdot \gamma$, is a measure-preserving bijection: it fixes the invariant start distribution and the equivariant transition and policy, and it

Table 4. Networks and decoders behind each figure and table. Original and replicate sweeps: same design, different seeds. Path-held-out decoder: whole trajectories and whole orbits held out of training. Orbit-held-out decoder: whole orbits held out; it can partly recognise paths and reads 0.55–0.58 on a fully folded code (Disambiguation accuracy, below).

figure or table	networks (n)	decoder or readout
Fig. 2a	original two-room networks (8 per maze)	room at unseen cells; position within a room (ridge)
Fig. 2b	original four-room networks (8)	as in (a)
Room decoding on held-out paths (Results)	replicate two- and four-room networks (8 per maze)	path-held-out (Table 5)
Table 2, Fig. 3c, d	replicate sweep	path-held-out
Fig. 3b	replicate sweep, one network per condition	none (rate maps)
Fig. 4a	replicate sweep, quarter-turn arena (8 per signal)	path-held-out, four-way
Fig. 4b–e	original sweep	none (geometry of the code)
Fig. 6a, b, d	original sweep, subsets (6, 6, 4)	orbit-held-out (a)
Fig. 6c	original two-room networks, fresh sample of trajectories (8 per maze)	repetition; position within a room
Full silencing in two rooms (Results)	replicate cue-sweep networks trained without a cue (6 per maze)	path-held-out room decoder
Fig. 7a	replicate sweep and learned-signal networks	ridge readout
Fig. 7b, c	replicate sweep, half-turn arena (10 per signal)	ridge readout; kernel theory
Fig. 7d–f	replicate two-room networks (8 per maze)	TD(0) value readout
Navigation (Results)	replicate sweep; 4 untrained networks	ridge steering readout
Other learners (Results)	GRU, LSTM, transformer (5 per signal)	path-held-out
Fig. 8a	original sweep, subsets (6, 6, 4)	none (single units)
Fig. 8b, c	city-block networks (8)	none (single fields)

leaves the sequence of observations and movement signals unchanged, because G-related positions are observationally identical and ϕ is G-invariant. Histories ending at x and at $g \cdot x$ therefore induce the same input sequence, so the network, a fixed function of that sequence, assigns them the same state, and the position posterior is constant on each orbit. No function of the state separates positions in one orbit, so a decoder that must choose among the $|G|$ positions of an orbit is bounded by $1/|G|$. This is the belief invariance of group-invariant partially observable processes (8, Lemma 1), specialised to a group under which every input is unchanged. The arena side is even, so no rotation fixes a cell and every orbit has exactly $|G|$ positions. The random walk used here (below) is the same at every position and heading, so it satisfies the condition on the policy.

Relation to state abstraction. The equivalence that defines the fold is set by the observation model, the head-direction signal included, and not by the symmetry of the underlying decision process alone. The half-turn is a symmetry of that process under every signal, so a state abstraction taken from it, such as a symmetry-induced homomorphism (30, 31), would predict a fold under *full* and *flip* as well, which the data reject. The relevant objects are the causal states of the observation process, the coarsest partition of histories that preserves their predicted futures (32). Recurrent networks trained to predict their next observation approximate them (33), and histories in one causal state are bisimilar (33, 34). The folded code is then the coarsest partition of histories that preserves the agent's future observations; for fully observed decision processes the corresponding object is the model-irrelevance abstraction (35). For *axis* and *const* in the C_2 arena the observations are exactly bisimilar (no position–heading pair has a current input that distinguishes it from its twin; Supplementary Information), which is why the fold there is exact.

Arenas. 18×18 MiniGrid arenas (324 positions) with landmark layouts of C_1, C_2 and C_4 rotational symmetry. The observational discriminability index (ODI; the Spearman correlation between heading-averaged observation distance and spatial distance across all pairs of positions) is *not* matched across the three arenas, and cannot be: symmetric landmarks make distinct positions observationally identical, so discriminability necessarily falls with symmetry order. Measured over all 324 positions, ODI $= 0.336$ (C_1), 0.288 (C_2), 0.262 (C_4). A fourth arena tests the result outside the square: an annulus (the 18×18 grid with a central hole) carrying the half-turn-symmetric landmark layout of the C_2 arena, with six networks per head-direction signal.

This does not confound the results, for two reasons. The central contrast, *axis* vs. *flip*, is made *within* one arena, on the same trajectories, so the observation stream and its ODI are identical between the two arms; the only difference is a 4×4 matrix on the heading block. And across arenas, ODI does not predict the readout: under *full*, disambiguation accuracy (orbit-held-out decoder, original networks) is 0.981, 0.978 and 0.984 for C_1, C_2, C_4, with the least discriminable arena scoring highest; three arenas are too few for a correlation, so we report the values only. Trajectories: 10,000 per arena, 200 steps, random start position and heading. The policy is a fixed-probability random walk (forward 0.60, turn left 0.15, turn right 0.15, stop 0.10), identical at every position, heading, and arena; nothing about it depends on which fundamental domain the agent occupies, which is what makes it measurable with respect to the agent's belief (Theorem). The

prediction horizon k is the number of future views predicted at each step: after each step the network is run k steps forward with a zero view and the true movement signals, and its predictions are scored against the next k views.

Head-direction signals. The heading block of the action vector is left-multiplied by a fixed 4×4 matrix: identity (*full*); $\frac{1}{2}(I + P^2)$ where P is the cyclic shift (*axis*); the block-averaging matrix pairing $\{E, S\}$ and $\{W, N\}$ (*flip*); and $\frac{1}{4}\mathbf{1}\mathbf{1}^\top$ (*const*). Column sums are preserved, so activation magnitude is unchanged. Entropies under a uniform heading prior are $2, 1, 1, 0$ bits.

Ensemble training. All networks sharing a computation graph (arena, signal and seed all leave it unchanged; the horizon k does not) are trained as a single batched job, which lowers every per-model matrix multiply to one batched multiply and makes the number of GPU operations independent of the number of models. Forward passes are bit-exact against separate runs; gradients are bit-exact with noise disabled and agree to 2.3×10^{-10} absolute with the production noise on (reordering of floating-point sums in batched multiplies). Per-model gradient clipping, the four-group RMSprop and the sum-of-per-model-means loss are all preserved; each is checked against the reference implementation by test.
Ensemble sizes differ across analyses, and Table 4 gives each.

Disambiguation accuracy. Hidden states are collected over 20,000 raw timesteps (no per-position averaging); repeated visits to one position therefore contribute multiple, slightly different rows. Orbits are $\{x, \rho^2 x\}$ under the $180°$ rotation (or the four-element C_4 orbits where stated); the arena side is even, so no rotation has an integer fixed point and every orbit has exactly $|G|$ cells (162 orbits at C_2, 81 at C_4). Each state is labelled by which position of its orbit the agent occupies, that is, by its fundamental domain, and the decoder must report that label. Labels are balanced. In the *orbit-held-out decoder*, every row from an orbit's positions is assigned to the same split, regardless of how many visits each contributed, and cross-validation splits hold out whole orbits, so the decoder must generalise the distinction between twins rather than memorise orbits or exploit a repeated visit to the same position appearing on both sides of a split. We report the accuracy of a logistic decoder, together with the R^2 of ridge decoders for the raw position and for the position within the fundamental domain. Codes that separate twins also carry the within-domain coordinate linearly ($R^2_{\text{domain}} = 0.916$ under *full* in the C_2 arena), and folded codes carry only it, so the check that a code still represents space is $\max(R^2_{\text{raw}}, R^2_{\text{domain}})$.
Holding out orbits is not enough when the code carries the identity of a trajectory. A trajectory stays in one half of the arena for long stretches (the agent is in the same fundamental domain 25 steps later with probability 0.797), and a recurrent state carries a fingerprint of its own history, so a decoder can learn which trajectory a state came from and carry that trajectory's usual label to held-out orbits. The standardiser is fit on the training fold only. The excess above chance comes from trajectory identity. Two codes with no position information at all, a one-hot trajectory label and a slowly drifting random vector per trajectory, read 0.771 and 0.871 under the orbit-held-out decoder. Untrained networks whose rotated hidden states are identical read 0.550–0.584. The *path-held-out decoder* therefore also holds out trajectories. Trajectories and orbits are each split into two halves; the decoder is trained on one trajectory half restricted to one orbit half and tested on the other trajectory half restricted to the other orbit half, and accuracy is averaged over the four assignments. Under the path-held-out decoder the two position-free codes read 0.500 and 0.483, and the untrained networks 0.488–0.500. Table 2 and Figs. 3 and 4a use the path-held-out decoder on the replicate sweep. The other disambiguation accuracies in the paper come from the orbit-held-out decoder on the original networks. On folded codes in the replicate sweep, that decoder reads 0.02–0.04 above the leak-free value (0.540 against 0.501 under *axis* in C_2), so we compare values only within one decoder, and the Results name the decoder behind every value.
The path-held-out decoder has low power for weak signals. On C_1 networks, where twins are decodable, we flipped each label with probability 0.446 in training and test alike, which leaves an expected accuracy near 0.55; it read 0.505–0.514 ($n = 4$ networks). A reading near 0.5 is therefore not, by itself, evidence that the fold is exact. Nor is a small departure from 0.5 evidence against it: on codes whose rotated states are identical, it reads between 0.477 and 0.507 across trajectory samples (untrained networks, and trained *axis* and *const* networks in both symmetric arenas). The evidence is the input identity. Under *axis* and *const* in the C_2 arena, a trajectory replayed from the position a half-turn from its start, with the same egocentric actions, produces identical views and identical encoded actions, and from the same initial state identical hidden states (largest difference 0.000 in every trained network, $n = 10$ per signal). The same holds for the half-turn in the C_4 arena ($n = 8$ per signal). The trajectory generator is itself symmetric under the half-turn (start cell and heading uniform, actions independent of position), so the states at a position and at its twin follow the same distribution, and no decoder can exceed chance in expectation. All networks are decoded on one sample of trajectories, so the one-sample tests against a floor take the network as the unit and do not include trajectory sampling.

Delayed reconstruction. To test an objective that needs memory without prediction, the target at step t is the view at step $t - d$ ($d = 1$ or 3, $k = 0$), so the network must hold a past view but never infers one it has not seen. Data, signals, seeds and training are those of the C_2 sweep ($n = 6$ per signal and delay); for $t < d$ the target is the first view. Map quality is the spatial representational similarity, the correlation between neural and Euclidean distance (Supplementary Information): 0.164–0.180 across signals and delays, against 0.243 for untrained networks in the same arena ($n = 5$). Disambiguation accuracy, with trajectories held out, is 0.500–0.506 for every signal at both delays.

Position relative to the start of an episode. Folded networks cannot report which fundamental domain the agent oc-

cupies, but they can carry a label that the symmetry leaves unchanged: whether the agent is still in the domain in which the episode began. We decoded $z(t) = \mathrm{domain}(t) \oplus \mathrm{domain}(0)$ with a logistic model under cross-validation grouped by episode and balanced class weights, and report balanced accuracy (chance $= 0.5$ regardless of z's base rate) as a function of the time since the episode started. Raw accuracy cannot be used here: under a random walk, $P(z = 1)$ rises from about 0.08 near episode start to about 0.46 by $100+$ steps purely from position continuity, independent of the hidden state, so a majority-class-only classifier scores about 0.93 at 0–10 steps with no information from the code. For $n = 10$ seeds each of *learned*, *axis* and *const* in the C_2 arena, balanced accuracy is nominally above chance (Wilcoxon signed-rank against 0.5, exact two-sided, uncorrected) at every time bin tested for all three signals, decaying from 0.66–0.74 near episode start toward the across-episode baseline (0.52–0.53) by $100+$ steps. The effect at the longest bin is markedly weaker for *learned* ($p = 0.049$, at the edge of detection) than for *axis*/*const* ($p \leq 0.0098$), though none actually falls to non-significance. The label z is unchanged when a trajectory is turned by a half-turn, so decoding it does not conflict with an exact fold: it reads whether the agent is still in the half of the arena where the episode began, a property of the path rather than of the absolute position. The network keeps this relative frame for a window after an episode starts, and the frame attenuates toward the steady-state floor rather than vanishing at the first step or never appearing at all.

Geometry of the code. For each network we averaged the population state at every arena cell and computed geodesic distances between those states on a nearest-neighbour graph. We asked which metric these distances reproduce better: the arena's own metric d_X, or the folded metric $d_{X/G}(x, y) = \min_{g \in G} \|x - g \cdot y\|$. Fit is Kruskal stress-1 at the best scale. In the C_2 arena, the *axis* code fits the folded metric at stress 0.190 and the arena metric at 0.468; the *full* code fits the arena at 0.097 and the folded metric at 0.485. Because the folded metric is never larger than the arena metric, a code whose distances stop growing at long range would favour it for a reason unrelated to symmetry; a sham group of the same order, a translation by half the arena, which shortens distances as the half-turn does, controls for this. The *axis* code prefers the true symmetry over the sham by 0.343, and the codes that separate twins prefer neither. Assigning each network to whichever of d_X, d_{X/C_2} and d_{X/C_4} fits best, 103 of 112 networks are assigned by their signal alone; eight of the nine exceptions are the *axis* networks in the C_4 arena, fitted better by X/C_4 although the decoder places them just under the half-turn ceiling (0.472). A larger group always compresses a code more, so the stress comparison is biased toward it in this nested case; no claim rests on the geometry where it and the decoder disagree. In a nonlinear Isomap embedding, a position and its twin lie at 0.17–0.18 of the distance between a random pair under *axis*, against 0.85–0.89 under *full* (Supplementary Information).

Single units. A unit's rate map is its mean activation at each arena cell. A field is a contiguous region above 50% of the unit's peak rate covering at least 3 cells; a unit is a place cell if its Skaggs spatial information is at least 0.3 bits per unit of activation and it has at least one field. Directional tuning is summarised by the strength of one-peaked (m_1) and two-peaked (m_2) modulation of the tuning curve over the four headings, the amplitudes of its first and second circular harmonics relative to its mean. The two-peaked share of a unit's tuning is $|F_2|^2/(|F_1|^2 + |F_2|^2)$, averaged over units with $m_1 + m_2 > 0.10$ (about 409 of 500 per network under *axis* in the asymmetric arena). The example rate maps of Fig. 3b come from one network of the replicate sweep per condition (seed 20): we stored the 40 units with the highest spatial information, computed for each the Pearson correlation r between its unsmoothed rate map and the map turned by a half-turn (by a quarter-turn for the fourfold value), and show the three place cells whose r lies closest to the median of the 40 (36 to 40 of the 40 are place cells in every condition). The symmetry index of Fig. 8a is the same correlation computed on smoothed maps of all units, so the two are not directly comparable.

Isotypic spectrum. Each unit's rate map is decomposed into the four characters of C_4; P_k is the normalised power in character k, $\sum_k P_k = 1$, and $P_1 = P_3$ for real maps. $\mathrm{RA} = P_0 - P_2$ and $\mathrm{odd} = P_1 + P_3$.

Readouts on frozen codes. Readouts are trained on the hidden states of frozen networks, standardised on training states only. *Goal transfer* (replicate sweep and learned-signal networks): the target is a Gaussian goal field (width 3 cells) on the within-domain coordinate, placed in one fundamental domain and zero in the others, and a ridge readout is trained on states from all domains. The spread into domain k is the mean prediction in domain k's goal region over the mean prediction in the goal's own region. Training and test states are disjoint in trajectories and in orbits (two halves of each, four assignments, three random splits). *Value learning* (replicate two-room networks, $n = 8$ per arrangement): a linear TD(0) value function on standardised states ($\gamma = 0.9$, step size 0.1 divided by the number of units), rewarded while the agent is in a 3×3 goal zone of room A, trained on $1{,}000$ trajectories presented once each. On 200 held-out trajectories, room discrimination is the AUC of value for the goal zone against the matching zone of room B, the within-room control is the AUC for the goal zone against cells of room A at least three cells away, and criterion is the first evaluation with AUC ≥ 0.8. The value carried into room B is $(\bar{V}_B - \bar{V}_{\mathrm{far}})/(\bar{V}_A - \bar{V}_{\mathrm{far}})$ over the two zones and the far cells. Interference is the peak within-room AUC minus its final value. The masking test skips every TD update at a room-B state. The lesion silences the heading block of the action input on a fraction p of steps ($p = 0.5$ or 1) at test time, in networks trained with an intact head-direction signal. Repetition and room decoding under full silencing (Results) use the replicate cue-sweep networks trained with no cue ($n = 6$ per arrangement).

Navigation in closed loop (replicate sweep): a readout steers the agent toward a goal cell. For each action, a ridge regression from the hidden state predicts γ^d, where d is the distance to the goal after that action ($\gamma = 0.9$), and the agent takes the action with the largest prediction. The view is shown on every sixth step, on which the agent stops; on the five steps between, the view is replaced by zeros, the input the networks receive during their own five-step rollout, and the readout chooses the action. Readouts are fitted on 90 random-walk trajectories and the same trajectories turned by each rotation and tested on 30 held-out ones. Each of 50 starts is run together with its three twin starts, from one shared initial state (200 trials per network, 60 decisions each). A trial is scored by where it ends: at the goal, at one of the goal's twins, or at none of them. With these symmetric starts a folded agent reaches the goal and its twins equally often by construction. On ground-truth codes, a readout of the true state ends at the goal in 0.97 of trials, a memoryless policy on the current view and the head-direction signal in 0.07, and random actions in 0.025. A second test steers the agent to one goal and then, with a second readout, to another; its design, ground truth and declared predictions are in the Supplementary Information.

Kernel theory. The kernel is linear on position-conditioned mean states (the mean hidden state at each cell visited at least 8 times), $K(x,x') = h(x) \cdot h(x')/D$, with positions drawn uniformly from cells whose orbit is complete. Eq. 4 of Canatar et al. (10) gives the generalisation error of kernel ridge regression from the kernel's eigenvalues and the target's power in each eigenmode. The ridge is set to 10^{-3} times the mean of $K(x,x)$ and the label noise to zero, so nothing is fitted. The share of variance that differs between twins is the part of the centred sum of squares of the mean states that lies outside the G-invariant subspace (each state minus the mean over its orbit), divided by the total; both are computed as products between two independent halves of the trajectories, so that noise in the mean states does not inflate the share. The number of training positions needed, P^*, is the value of P at which the relative error first reaches 0.1 (log–log interpolation). The gain from imposing invariance on a code is P^* for the code divided by P^* for the same code with each mean state replaced by the average over its orbit. Measured curves draw $P = 2$ to 512 training cells with replacement (20 draws per P). A fit passes when the mean absolute $\log_{10}$ difference between the predicted and the measured curve is smaller than for each of 20 shuffles of the eigenvalues. At a near-ridgeless 10^{-5} the theory passes 495 of 592 fits, and 206 of 296 for the goal confined to one domain. For TD, the projected fixed point is the least-squares TD solution on the states of the TD readout.

Offline replay. Offline rollouts follow Levenstein et al. (7): the network is driven by its own predicted observation and a random-walk action sequence. Coverage and sequenceness decode position with a ridge readout fit on waking states, in raw arena coordinates and, separately, in the fundamental domain; coverage is the number of distinct cells a rollout visits over the number a waking path of equal length visits. These rollouts pass the heading to every network as a one-hot code, including networks trained on an encoded heading. The replicate analysis passes the heading through each network's own head-direction signal, decodes the within-domain cell with a multinomial classifier and the domain with a binary classifier, both fit on waking states of training paths, and divides the within-domain coverage of offline rollouts by that of held-out waking paths; domain confidence is the mean $|p - 0.5|$ of the domain classifier. Passing the heading as a one-hot code instead changes the mean within-domain coverage of a signal by at most 0.010 and its mean domain confidence by at most 0.007 (single networks: 0.023 and 0.015). Before any network was scored, a rollout whose within-domain coverage is below 0.25 of the waking coverage was declared a rollout that does not move. The ridge readouts were also run on the replicate networks and scored against decoded held-out waking paths. The reference waking path of the first replay analysis is a single path, the same for every network.

Room mazes: construction. Two (or four) closed 6×6 rooms with identical landmarks open onto one corridor. Rendering every room-interior cell and heading and comparing the images across rooms gives zero mismatch (0 of 144 pairs with two rooms, 0 of 432 with four), so inside the rooms every view occurs at the matching spot of every room. A network's repetition score is the median, over its 100 most variable units, of the Pearson correlation between a unit's mean activity map in one room and in the other, over matched room-interior cells visited at least five times in both rooms. Room identity is decoded from room-interior states only. Every translation network differs from every rotation network in repetition, room decoding and within-room R^2 (0.95 and 0.79; Mann–Whitney $p = 1.6 \times 10^{-4}$, the floor for 8 against 8).

Room mazes: the corridor. The corridor breaks the symmetry in two ways, so the room mazes test the principle empirically rather than instantiating the theorem. First, every door opens onto one shared junction, and a room at either end of the junction's span sees it truncated on one side; once the forward view reaches the junction, the difference is visible (largest single-channel pixel difference $5/255$). Second, the two-room corridors differ in length: the translated rooms are joined by a direct link of 27 cells, and the rotated rooms by a link of 57 cells that runs around three edges of the arena. A longer corridor could let memory of the transit, rather than the head-direction signal, carry the room's identity into the room. Decoding by time since entry argues against this (Table 5). Under translation, decoding is highest just after entry and decays to chance within about ten steps. Under rotation it is lowest just after entry and rises to its ceiling, the opposite of what a corridor memory predicts and what a network that needs a few steps of movement to resolve the room would show. The corridor trace makes the translation value a slightly conservative estimate of the fold and does not touch the rotation result.

Room mazes: decoding the room. A decoder trained and tested on states from the same paths can win by recognising the path, because a path stays in one room for long stretches. The main text therefore uses replicate networks ($n = 8$ per maze) decoded with whole paths held out and local position balanced between rooms, and compares them with reference codes that carry no room information, analysed on the same states (Table 5). After the entry bin, the two-room translation networks still exceed a memoryless code ($+0.044$ in steps 3–9 and $+0.016$ from step 10; Wilcoxon $p = 0.0078$ each), so a small trace of the last entry persists. In the four-room maze, a declared test, held-out-path decoding within 0.25 ± 0.01, was inconclusive (0.238, every network below 0.25), because a code with no room information reads 0.235 ± 0.015 over 32 samples of paths; we therefore compared the networks, post hoc, with reference codes. In the first steps after entry the four-room networks decode the room above a memoryless code in every network (0.273 against 0.236).

Graded-cue sweep. Room B's views carry a tint ϵ against fixed per-pixel Gaussian noise ($\sigma = 0.05$), so a single view separates the rooms with $d' = \epsilon\sqrt{N}/\sigma$ for N view pixels; eight nominal levels span $d' = 0$ to 3, with $n = 8$ networks per level. Training reads views from a store that holds them as 8-bit integers ($\mathrm{round}(255\,o)$), so the tint a network trained on is $\mathrm{round}(255\epsilon)/255$ and the effective training cue is $d'\,\mathrm{round}(255\epsilon)/(255\epsilon)$: the nominal levels 0, 0.43, 0.86, 1.29, 1.71, 2.14, 2.57 and 3.00 train at 0, 0, 0.95, 0.95, 1.90, 1.90, 2.85 and 2.85. Evaluation renders views in floating point with the nominal tint. We report the sweep against the effective training cue, and the paired difference between the two nominal levels of each effective level isolates the evaluation cue. A replicate sweep (6 seeds per nominal level, 12 networks per effective level) was decoded with whole paths and local cells held out. To separate learned from input-driven room information, networks trained with no cue were evaluated with the cue at $2/255$, and networks trained with the cue at $2/255$ were evaluated without it.

CA1 reanalysis. From Hockeimer et al. (11) (JHU repository doi:10.7281/T15HMQD4) we took, for each place field, its per-visit firing rates split by travel direction, and formed a directional index $\mathrm{DI} = (r_+ - r_-)/(r_+ + r_-)$ with $+/-$ the two directions of the field's alley (E/W for horizontal alleys, N/S for vertical); fields with fewer than three visits in either direction were dropped. The primary statistic is the Pearson correlation over within-cell same-orientation field-DI *pairs* (a cell's repeating fields lie in translation-related alleys). The shuffle null permutes DI values across all repeating fields *within an orientation*, breaking the assignment of DIs to cells while preserving the marginal DI distribution, and is recomputed 2000 times. This null, however, also destroys cell-level directionality, so we ran two further controls that hold it fixed. First, the same correlation on *non-repeating* cells' same-orientation fields (which the account does not predict should correlate). Second, a directionality-matched scramble that preserves each cell's per-orientation mean DI and resamples the within-cell residuals from the global pool, isolating the between-cell (global-directionality) contribution; the directional index is substantially a between-cell property (mixed-effects ICC 0.31), and this scramble reproduces (indeed exceeds) the observed correlation. The observed correlation is therefore consistent with the account but not separable from a conjunctive place-by-direction code in these data.

Statistics. Contrasts between groups of networks are two-sided exact Mann–Whitney U tests on per-network values, and paired contrasts within networks are exact Wilcoxon signed-rank tests, unless stated otherwise. Other tests are named where they are used: one-sample t-tests against a chance floor, an F-test across three arenas for the orbit variance under silencing, and Spearman or Pearson correlations. The planned contrasts are *axis* vs. *flip* within each arena, at $n = 10$ per cell for C_1 and C_2 and $n = 8$ for C_4; at $n = 10$ vs. 10 the exact two-sided floor is $2/\binom{20}{10} = 1.1 \times 10^{-5}$, and a p-value at that floor indicates complete separation (every network in one group above every network in the other). Bonferroni correction across the three planned arena contrasts is applied where stated. Secondary experiments carry their own n: the prediction-horizon and noisy-signal sweeps use $n = 6$ per cell (floor 0.00216), the isotypic re-run $n = 5$ (floor 0.0079). Accuracies of folded codes are compared to their $1/|G|$ floor by a one-sample t-test. No results were excluded; models failing a criterion are reported as such rather than dropped.

The exact ceiling. Views and movement signals are deterministic functions of the state (cell, heading and room) and the action, so the posterior over states given the history of inputs can be computed exactly by a forward filter over the enumerated states, under the random-walk policy and the uniform start of the training data. The ceiling for a label (the twin member, or the room) is the expected largest posterior probability of the label at each step, averaged over $2{,}000$ sampled paths of 200 steps. The filter was validated on a toy arena with an exact half-turn symmetry and no heading signal (ceiling 0.500 at every step) and on one with a unique landmark (ceiling 1.000). Training added input dropout and hidden-state noise, which the filter does not model, so the ceiling is an upper bound.

Retention and its lag. A room step is silent when the rendered view and the movement signal at the state equal those at the matched state of the other room; the lag is the number of steps since the last step that was not silent. The corridor arms render identically in both arrangements and so count as silent: the lag clock starts at the junction view, not at room entry. Retention at each lag is room decoding with the path-held-out decoder, on steps where the ceiling is at least 0.99, scaled between a memoryless code run through the same pipeline and the ceiling; the half-life is the model-free lag at which this scaled value first falls below one half. The predictive worth of remembering the room is the expected reduction of the squared error of the next $k = 5$ views when the Bayes-optimal predictor knows the room, against one that holds the room at its prior on silent steps, conditioned on the actual future movement inputs as in the network's own rollout; it is reported relative to the networks' converged loss and is an

Table 5. Room decoding in the room mazes, by code and decoder (two rooms: chance 0.5; four rooms: chance 0.25). Replicate networks, $n = 8$ per maze, unless marked as original.

maze	code	decoder	room decoded
two rooms, translation	networks	paths held out, all room states	0.497
	networks	paths held out, steps 0–2 / 3–9 / 10+ after entry	0.645 / 0.506 / 0.477
	memoryless code	paths held out, steps 0–2 / 3–9 / 10+	0.473 / 0.462 / 0.461
	networks	two decoders that share paths	0.512, 0.509
	original networks	nearest neighbour at seen positions	0.515
two rooms, rotation	original networks	steps 0–1 / 2–4 / 5–9 / 10+ after entry	0.71 / 0.92 / 0.98 / 0.99
four rooms, translation	networks	paths held out	0.238
	memoryless code	paths held out	0.228
	path-identity code	paths held out	0.244
	networks / original	generalising decoder	0.260 / 0.262
	networks / original	nearest neighbour	0.264 / 0.268
	memoryless code	generalising / nearest neighbour	0.247 / 0.261

upper bound.

The room-memory subspace. For each network, paths that passed the junction were paired with the same action sequence entering the other room; the room-difference subspace is spanned by the principal directions holding 95% of the variance of the difference between paired states over lags 0 to 30. Ablation projects this subspace out of the hidden state at every step from the last informative step on, inside the network's own forward pass; the control projects out a random subspace of equal dimension and variance, orthogonal to it. The room decoder is trained on intact states and applied unchanged.

Retrained controls. Two-room translation networks were retrained from the initial weights of the original networks (seeds 20 to 25), each arm differing from a same-seed baseline in one change, as verified by unit tests: (i) an added loss, weight 1, on a linear readout of the room from the hidden state at every room-interior step; (ii) training paths and truncated backpropagation of 600 steps instead of 200, with 96 loss positions per path instead of 32, so that loss positions per step are unchanged. Training ran on one L40S GPU; all arms were evaluated on the same 200-step data.

Tests behind the Results. The main text gives one number per claim; the tests are collected here, in the order of the Results. *Matched pair:* in the half-turn arena every *flip* network scores above every *axis* network (Mann–Whitney $U = 0$, $p = 1.1 \times 10^{-5}$, the floor for 10 against 10; $p = 3.2 \times 10^{-5}$ after Bonferroni correction across the three arenas); in the asymmetric arena the two do not differ ($\Delta = -0.001$, $p = 0.39$). *Ceilings: const* in the quarter-turn arena reads 0.248 against the $1/4$ ceiling (one-sample $t(7) = -1.36$, $p = 0.22$, $n = 8$). Read with a half-turn decoder, the *const* code of that arena seems to carry no spatial information, yet position within one quarter of the arena is decodable from it ($R^2 = 0.646$, minimum 0.535, $n = 4$ networks of a first quarter-turn sweep). *Twins as one point:* the cosine between twin states is higher under *axis* than under *flip* in every network ($U = 100$, $p = 1.1 \times 10^{-5}$). *Silencing:* silencing sets the heading block of the action input to zero, at test time, in networks trained with *full*; the unsilenced network is labelled intact in Fig. 6. The map of the folded arena is measured as the variance of each unit explained by the orbit the agent is in, averaged over units, and reported as a change from the unsilenced network. It falls by 37.6%, 30.9% and 38.1% in the asymmetric, half-turn and quarter-turn arenas; the falls are not identical ($F = 10.78$, $p = 0.002$) but are not ordered by symmetry (asymmetric against quarter-turn, $U = 15$, $p = 0.61$; half-turn against quarter-turn, $U = 24$, $p = 0.010$). Disambiguation accuracy under full silencing is higher in every asymmetric-arena network than in every symmetric-arena network ($U = 60$, $p = 2.5 \times 10^{-4}$, $n = 6$ against 10). The partial-silencing panel (Fig. 6c) uses the original two-room networks on a fresh sample of trajectories, on which their intact rotated-room repetition is -0.198 (against -0.080 in Fig. 2a). With the signal silenced on every step in two rooms, a value readout still discriminates positions within a room (area under the ROC curve 0.943). *Learning:* under rotation the TD readout reaches criterion after 3.5 episodes on average; under translation no network reaches it ($0/8$). Its within-room control reaches 0.976 and 0.997 under translation and rotation. The projected TD fixed point predicts the value carried into room B (0.999 against a measured 1.001 under translation; -0.125 against -0.092 under rotation). The interference fall differs between arrangements (Mann–Whitney $p = 0.0047$, $n = 8$ each), and blocking room B reduces it in every translation network (Wilcoxon $p = 0.0078$). The effect of full silencing on room discrimination (rotation 0.521, translation 0.499) separates the two arrangements completely ($p = 1.6 \times 10^{-4}$, $n = 8$ each); silencing half of the steps has a smaller effect (rotation 0.938 on average: seven networks end above 0.99, and one ends at 0.52 after reaching criterion at episode 5). *Spectrum: axis* needs fewer training positions than *full* in every network ($p = 1.1 \times 10^{-5}$); *flip* needs 35.8 (predicted 36.6) and *const* 29.6 (predicted 29.9). *Navigation:* the *full* agent beats untrained networks (0.904 against 0.50, $n = 10$ against 4, $p = 0.002$); under *const* in the quarter-turn arena the agent ends at each of the four twins equally often (0.123 each). *Single units:* under *axis* in the half-turn arena 93.1% of units have a symmetry index above 0.8 (median 0.94); under *full*, 54.3% score below 0.2, so the index can return the negative answer. *City-block maze:* the same-

orientation correlation exceeds the orthogonal one in all 8 networks (Wilcoxon $p = 0.0078$).

Data and code availability

The source data underlying every figure and table are provided as comma-separated files in the code repository; an index in the same folder maps each analysis to its data file and script. All code, the analysis pipeline that regenerates those files from trained checkpoints, and the figure-generation scripts are openly available at https://github.com/RavulapalliManas/Thesis_work, and will be archived on Software Heritage on acceptance. The test suite pins the claims that fail silently: the bit-exactness of the batched ensemble against per-model training, that the $k = 0$ target is exactly the anchor observation, that a synthetic C_2-folded code reads chance disambiguation accuracy while a code with no spatial signal is distinguishable from it, and that the isotypic identity $\mathrm{RA} = P_0 - P_2$ holds. Trained checkpoints of the replicate sweep and the raw trajectories are available from the author on request. Every published number is reproducible from the committed data files without them.

Author contributions

M.V.S.R. designed the study, implemented the models and analyses, ran the experiments and wrote the manuscript.

Competing interests

The author declares no competing interests.

Use of AI tools

The author used Claude (Anthropic), a large language model, to write and review analysis and figure code, to draft and edit the text of this manuscript, and to check numbers and citations against their sources. The author designed the study, directed every analysis, reviewed the code and the text, and takes full responsibility for the content. Every number reported here is produced by the archived analysis scripts.

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